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Does Corruption Hurt Employment Growth of Financially Constrained Firms More?

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Abstract

Payments of bribes and the expenses incurred on rent-seeking activities impose a significant financial burden on private firms, which is compounded when they do not have enough funds of their own or find it costly to borrow externally. This paper hypothesizes that financial constraints magnify the harmful effects of corruption. It applies this idea to the impact of corruption on employment growth among private firms. Using firm-level survey data for 109 countries, the analysis finds that corruption has a much larger negative impact on employment growth for firms that are financially constrained compared with firms that are not financially constrained. For the baseline specification, a one standard deviation increase in the bribery rate brings about a decline in the annual growth rate of employment of financially constrained firms that is 2.3 percent greater than that for firms that are not financially constrained. This is a large difference given that the mean employment growth is about 5.1 percent. The results show that corruption "sands the wheels" at high levels of financial constraint and "greases the wheels" of an otherwise slow bureaucracy at low levels of financial constraint.

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1. Introduction

Corruption imposes significant monetary costs on the private firms. These costs include both the direct transfer of money (bribe paid) and the indirect costs incurred through the devotion of resources to rent-seeking activities. The loss of funds to expenses related to corruption is especially burdensome for firms that do not have enough internal funds or find it difficult to borrow externally. Thus, financial constraints can magnify the harmful effects of corruption. The present paper applies this idea to the impact of corruption on employment growth among private firms. We do so for a measure of overall corruption and two measures of petty corruption that arise in conducting a specified list of public transactions. The empirical exercise confirms that corruption hurts employment growth more for firms that are financially constrained compared to firms that are not financially constrained. In fact, for firms below a critical level of financial constraints, higher corruption is associated with higher employment growth, suggesting that corruption "greases the wheels" for such firms. The opposite holds for firms above a critical level of financial constraints, suggesting that corruption "sands the wheels" for these firms.

The interplay between corruption and access to finance has not been explored in the literature. In fact, there is very little discussion on factors that magnify or mitigate the effects of corruption. The literature is almost exclusively focused on the overall effects of corruption. Two contrasting views have emerged. First, the dominant view is that corruption is detrimental to economic growth and the development of the private sector (Shleifer and Vishny 1993; Svensson 2005; Fisman and Svensson 2007). In short, corruption "sands the wheels" of the economy. The alternative view is that corruption can have positive effects by catalyzing an otherwise slow and unresponsive bureaucracy (Leff 1964; Lui 1985; Dreher and Gassebner 2013). This is the "grease the wheels" view of corruption. The literature treats these two views as mutually exclusive, such

that corruption is either bad or good. With a few exceptions, the idea that corruption can hurt some groups and benefit others at the same time depending on their underlying conditions (such as access to finance) has not been explored.

Another important gap in the literature is the impact of corruption on job creation or employment growth. Fisman and Svensson (2007) estimate the impact of corruption on the sales growth rate but they do not consider the impact on employment growth. Other studies show that corruption adversely affects macroeconomic growth (Mauro 1997), productivity and investment (Mauro 1995, 1997; Keefer and Knack 1995; Tanzi and Davoodi 1997), foreign investment (Alesina and Weder 1999; Wei 2000), innovation (Gupta et al. 1999), and the quality of education and income equality (Transparency International 2007).

The are several ways identified in the literature in which corruption affects the private sector. First, payment of bribes involves a direct loss of revenue to the private firms. This deprives the firms of investible surplus. Second, by acting as a tax on economic activity, corruption distorts incentives to invest and innovate. Corrupt public officials tend to target the more profitable firms as these firms have higher ability to pay bribes. Thus, corruption dampens incentives for firms to improve efficiency and become more dynamic. Third, corruption can lower firm performance because it causes talent (entrepreneurial, managerial, and labor) and resources to shift away from their most productive use to unproductive rent-seeking activities (Murphy et al. 1991; Shleifer and Vishny 1993; Dal Bo and Rossi 2007). This occurs because corruption raises the returns to rent-seeking activities vis-à-vis productive activities. Fourth, corruption can undermine management quality because firm owners may prefer to hire managers with local ties instead of focusing on human capital and managerial skills. Desire to maintain secrecy over illegal transactions can result in excessive centralization in the decision-making process, create mistrust between co-workers,

tilt workforce composition towards more administrative staff to deal with corruption issues, and weaken incentives for monitoring the production process to minimize the financial information on which public officials can extract bribes (La Porta et al. 1997; Athanasouli and Goujard 2015). Fifth, uncertainty associated with bribes, arising from the demand of different bureaucrats for bribes on the same projects either causes firms to adopt easily reversible suboptimal production methods, the so-called "fly-by-night" production technology, or completely shut down their operations (Shleifer and Vishny 1993; Choi and Thum 1998). Last, corruption retards growth and productivity by decreasing government revenue and the quality of public services (Tanzi and Davoodi 1997).

Another strand of the literature documents the critical role played by access to finance in firms' growth and development. Firms with easier access to finance can take better advantage of available opportunities to grow (Rajan and Zingales 1998; Fisman and Love 2007). One reason for this is that the growth process often requires substantial amount of expenditure on physical capital, innovation activity, adoption of new technology, and so on. Better access to finance allows firms to innovate (Girma, Gong, Gorg 2008), invest in factors such as labor-saving technology (Fafchamps and Schundeln 2010), and in physical capital. Literature has shown that access to finance is particularly important for small firms, as they can lack the internal resources to innovate (Beck and Demirguc-Kunt 2006; Schneider and Veugeler 2010). While literature has shown how external finances can aid in firms' growth and development, the role of finance in alleviating some of the harmful effects of corruption has not been explored.

Our motivation for exploring the corruption-finance nexus is simple. As the review of the channels through which corruption impacts the private sector indicates, corruption imposes significant expenses on the firms. These expenses include both the amount of bribes paid (direct

cost) and the expenses incurred for the rent-seeking activity (indirect cost). As we discuss in the next paragraph, just the direct costs can be substantial and larger than the expenses on electricity. Meeting these expenses is more costly for firms that are financially constrained. For instance, we argued above that corruption hurts firms because it diverts resources from productive activity to rent-seeking activity. Firms that have easy access to finance can compensate for the loss of funds by borrowing. As financial constraints rise, the option of borrowing becomes increasingly more costly and less feasible.

Before going to the formal empirical analysis, we provide some descriptive analysis about the financial burden that corruption imposes on firms. The Enterprise Surveys (ES) data that we use for our empirical analysis provides some insights into the magnitude of the bribe payment vis-à-vis other cost elements. We focus here on firms that report paying bribes (about 20 percent of the sampled firms), as the role of access to finance in our model is conditional on payment of bribe. For these firms, Figure 1 shows our main corruption measure, that is the amount of bribes firms pay public officials to "get things done" as a percentage of firms' annual sales. For comparison, the figure also shows the electricity costs (as percentage of annual sales) for the same firms. All averages shown are median values. Reported bribes are higher, equaling 2.77 percent compared with 0.13 percent of electricity cost. We confirm that the qualitative nature of the finding reported here holds even when we exclude from the sample firms with no electricity cost. Thus, the direct cost of corruption (bribes paid) is significant and likely to put a strain on firms' financial condition. Information on the indirect monetary cost of corruption is not available.

Another piece of evidence comes from firms' perception of access to finance as an obstacle to firms' current operations. If corruption imposes significant costs on firms, then firms that pay bribes should find access to finance as a bigger obstacle to their current operations than firms that

do not pay bribes. Of course, this positive relationship between bribe payment and the severity of access to finance as an obstacle may be muted in the data because the more financially constrained firms are less able and therefore less likely to pay a bribe. Despite this, we do find evidence of a strong positive correlation between the two. Figure 2 illustrates the point graphically.

The present paper contributes to the literature discussed above in several ways. First, we explore the impact of corruption on job creation, a hitherto neglected issue as far as the effects of corruption are concerned. Second, we examine the role of financial constraints in determining whether corruption hurts or benefits the firms, and by how much. Thus, we bring together two contrasting models of corruption - "grease the wheels" and "sands the wheels" – together under one unified model. Third, as Fan et al. (2009) and Svensson (2003) note, most previous studies have used perceived corruption indices that rely on the aggregated perceptions of businesspersons or country experts, many of whom may have formed impressions—perhaps subconsciously—based on common press depictions of countries or conventional notions about what institutions or cultures are conducive to corruption. Thus, the use of these indices raises concerns about perception biases. The *Manual on Corruption Surveys* prepared by the UN (UNDOC, UNDP and UNODC-INEGI 2018) argues that corruption measures that are based on the actual experience of individuals with corruption (obtained from survey data) are far superior to the corruption measures based on opinions and perceptions of the experts. Thus, we depart from the literature by using firms' experience with corruption instead. We do so by employing measures of the overall corruption and petty corruption.

Fourth, we pay due attention to endogeneity concerns. We do so in several ways. For corruption faced by the firm, we proxy this by the average level of corruption experienced by all other firms (that is, excluding own firm) in the country-industry cell to reduce endogeneity

concerns. We do the same for a measure of financial constraints of the firm. We control for all time invariant country specific factors (country fixed effects) and so our results are immune from endogeneity problems that plague much of the cross-country evidence on corruption issues. We go beyond and argue that if our results our indeed causal and not spuriously driven, we should find that better access to finance alleviates the harmful effects of corruption more in countries where the enforcement mechanisms for implementing laws and rules are weak. The assumption here is that weak implementation of the laws allows public officials to favor those who have the money to pay bribes, paving the way for the corruption-finance nexus in the sense discussed above. If our results are indeed picking up the effect of the corruption and finance nexus on employment growth, we can expect them to be stronger when law enforcement is weaker. However, if our results are spuriously driven, there is no reason to expect the corruption-finance relationship to vary with the quality of law enforcement. Our results easily pass this test. To further raise our confidence against endogeneity concerns, we contrast the effects of corruption with other aspects of the business environment that are correlated with corruption but there is no reason to expect their effects to depend on access to finance. These business environment elements include power outages, burdensome regulations, and enforcement quality proxied by how well the courts function.

Fifth, we use micro-level data which allow us to capture within-country variations in the level of corruption and regulatory burden. Most of the existing studies use macro-level data which cannot capture these variations. As Svensson (2003) notes, firms facing similar institutions and policies may still end up paying different amounts in bribes for the same amount of services received. This is particularly important when considering the role of financial constraints since financial constraints tend to vary significantly across firms within the country.

Our results show that the effect of corruption on firms' employment growth depends on their financial condition. That is, as financial constraints faced by a firm rise, corruption's impact on employment growth becomes significantly more negative. Figure 3 illustrates the point graphically for our overall corruption measure and Figure 4 does the same for petty corruption. Based on our most conservative baseline results, for a one standard deviation increase in overall corruption, the decline in the annual growth rate of employment is 2.3 percentage points more for the most financially constrained firms than for the least financially constrained firms. The corresponding figures for the incidence of petty corruption equal 2.8 and 3.2 for the depth of petty corruption. These differences in corruption's impact on employment growth are economically large given that the mean level of employment growth in our sample is about 5.05 percent.

In addition to showing heterogeneity in corruption's impact, our results also have important implications for the overall impact of corruption. We show that when firms face above a critical level of financial constraints, corruption is associated with a significantly lower employment growth rate. In contrast, when firms face below a critical level of financial constraints, corruption is associated with a significantly higher level of employment growth rate. Thus, depending on firms' financial constraints, corruption "sands the wheels" for some firms and at the same time "greases the wheels" for others.

The plan of the remaining sections is as follows. In Section 2, we discuss the data and estimation methodology. The baseline regression results for overall and petty corruption are discussed in Section 3. In Section 4, several robustness checks are provided. The section also includes checks against endogeneity concerns. The concluding section summarizes the main findings of the paper and suggests scope for future work.

2. Data and Main Variables

2.1 *Data description*

The main data source we use is firm-level survey data collected by the World Bank's Enterprise Surveys (ES). In some specifications, we complement the ES with World Development Indicators, World Bank.

The ES are nationally representative surveys of the non-agricultural and non-financial private economy. A common sampling methodology – stratified random sampling – is followed in all the surveys along with a common questionnaire.[1] The sample for each country is stratified by industry, firm-size, and location within the country. Weights are provided in the survey and used in all our regressions so that the sample is representative of the target population.

Throughout, the sample used is a pure cross-section in that each country and firm is included only once. The most recent round of ES available for the country is used. The firms were surveyed between 2006 and 2018. For our baseline results for the main corruption measure, the sample consists of 41,203 firms. Of these, 21,823 belong to the manufacturing sector and the remaining 19,380 are in the services sector. The firms are spread over 124 countries and 7 industries (2-digit ISIC rev. 3.1). For later reference, the baseline sample includes 705 country-industry pairs. The sample size drops slightly to 41,117 when using the petty corruption indicators due to missing data.

2.2 *Estimation methodology*

The baseline regression exercise involves estimating the following equation:

[1] Details of the sampling methodology and other related information are available at www.enterprisesurveys.org.

$$Y_{ijk} = \alpha + \beta_1 \text{Corruption}_{-i,jk} + \beta_2 \text{Financially Constrained}_{-i,jk}$$
$$+\beta_3 \text{Corruption}_{-i,jk} * \text{Financially Constrained}_{-i,jk}$$
$$+ \text{CFE}_j + \text{IFE}_k + \text{Firm Controls}_{ijk} + u_{ijk} \qquad (1)$$

where subscript *i* denotes the firm, *j* the country and *k* the industry (2-digit ISIC rev. 3.1) to which the firm belongs. *Y* is the annual growth rate of employment over the last three years; *Corruption* denotes the level of corruption experienced by the firm; *Financially Constrained* is a measure of financial constraints faced by the firm; *CFE* denotes dummy variables for the country where the firm is located (country fixed effects) and *IFE* denotes dummy variables for the industry (industry fixed effects). *Firm Controls* includes various firm-level controls that are discussed in detail below and *u* is the error term. In the robustness section (Section 4), we include several additional controls which include interaction terms between firm and country characteristics and our measures of corruption and financial constraints.

The estimation methodology used is Ordinary Least Squares (OLS) with country and industry fixed effects employed throughout. Thus, the identification of our main results comes from variation within- rather than across countries. All the regressions use robust standard errors and are clustered at the country times industry level. We pay due attention to endogeneity concerns which we discuss in detail in Section 4.

A formal definition of all the variables used is provided in Table A1 in the online appendix. Summary statistics of the variables used in the baseline regressions are provided in Table A2 in the online appendix. The correlations between these variables is provided in Table A3 in the online appendix.

2.3 *Dependent variable*

The dependent variable is job creation or the annual growth rate of employment at the firm over the last 3 years. Employment here incudes all full-time permanent workers. The growth rate is computed is two different ways. First, we compute the average annual change in employment and divide it by the average level of employment in the initial and final year. The ratio is multiplied by 100 to arrive at *Employment Growth*. By construction, the variable is bounded between plus/minus 100 and so is less affected by extreme values. In our baseline sample, the mean value of *Employment Growth* equals 5.05 and the standard deviation equals 16.20.

Our second measure of employment growth equals the traditional change in log of employment between the initial and final year (log difference) and divided by the number of years. The ratio is converted to percentage by multiplying by 100 (*Employment Growth* 1). In our baseline sample, the mean value of the variable equals 5.30 and the standard deviation is 18.00. Our results for the interaction term between corruption and financial constraints hold for both the employment growth measures. This holds for the baseline results discussed in Section 3 and all the robustness checks discussed in Section 4. For brevity, only the baseline results are shown using the alternative employment growth variable.

2.4 *Main explanatory variables*

2.4.1 *Overall corruption*

The ES asks firms the amount of bribe firms like itself typically pay to public officials to "get things done". The motivation for the question is that firms are most likely to report their own experience with paying bribes. Thus, the responses to the question acts as a good proxy to the experience of the firm rather than the subjective opinions of experts. The resulting indicator is also

used as a variable in other studies including Fan et al. (2009), Duvanova (2014), and Amin and Soh (2020).

Firm-level responses to the bribery question cannot be used directly in our regressions as they are likely to be endogenous to the firms' growth experience (reverse causality problem) and firm characteristics (omitted variable bias problem). For instance, large firms are more visible and have higher ability to pay bribes. Thus, these firms may be targeted by corrupt public officials. At the same time, growth rate is also known to vary systematically with firm-size. One solution suggested in the literature is to proxy the level of corruption faced by a firm by the average level of corruption experienced by all other firms (that is, other than the firm in question) in the country-industry cell.[2] Using country-industry cell average also helps to control for potential measurement error if some firms choose not to respond or misreport the regulatory burden (Pounov 2016). Thus, we define our main explanatory variable, *Overall Corruption*, as the average of the bribe rate reported by the firms "to get things done" where the average is taken over all firms in the country-industry cell excluding the firm in question. For later reference, the term "CI avg." denotes the cell average at the country-industry. To ensure adequate thickness within the cells, all cells with fewer than 5 firms are excluded from the sample. As mentioned above, there are 712 country-industry cells in our baseline sample for overall corruption. In this sample, the mean value of *Overall Corruption* equals 1.18 and the standard deviation equals 1.81.

2.4.2 *Petty corruption*

The ES also contains information on instances of corruption that firms experience in soliciting the following: obtaining electricity connection, obtaining water connection, obtaining construction

[2] Reverse causality from a firm's own growth rate to the bribe payments experienced by the other firms in the country-industry cell is highly unlikely, although it cannot be ruled out completely (see for example, Aterido et al. 2011).

permit, obtaining import license, obtaining operating license, and inspections or meetings with tax officials. Based on this information, the ES compiles two separate measures of petty corruption. The first measure is the incidence of petty corruption, defined as a dummy variable equal to 1 if a firm experienced a bribe payment or request in one or more of the six transactions listed above and 0 if otherwise. We use average value of the dummy as our next corruption measure, in which as above, the average is taken over all firms in the country-industry cell other than the firm in question (*Petty Corruption Incidence*). Thus, the proportion of firms in the country-industry cell that experienced an incidence of petty corruption is used as a proxy for petty corruption experienced by the firm in question. The second measure of petty corruption compiled by the ES is the depth of petty corruption, defined as the proportion of the six transactions for which the firm experienced a bribe payment or request for one. Similarly, we use average values of the depth of petty corruption as our third corruption measure, where the average is taken over all firms in the country-industry cell other than the firm in question (*Petty Corruption Depth*).[3]

In our baseline sample, the mean value of *Petty Corruption Incidence* equals 0.18 and the standard deviation is 0.17. The corresponding values for *Petty Corruption Depth* are 0.14 and 0.15 respectively.

2.4.3 *Validating the corruption measures*

The overall corruption measure aims to capture all types of corruption that the private firms experience in running their business. Thus, it is ideally suited for understanding the overall impact of corruption on job growth. However, the overall corruption measure is based on an open question

[3] The two measures of petty corruption are available for only those firms that solicited the public services listed above or were inspected by tax officials. For the bribery depth measure and following the ES methodology, a refusal to answer a question on whether bribes were requested or expected is considered as an affirmative answer.

on how much bribes firms pay to "get things done". While the measure has been used in the literature, there may be some concern since it is not clear what types of corruption experiences firms have in mind in responding to the question. One way to address such concerns, at least to some extent, is see how well the overall corruption measure correlates with the petty corruption measures. Since petty corruption is part of overall corruption, we should find that the two are positively correlated. Of course, the correlation is unlikely to be perfect since petty corruption is only a fraction of overall corruption, and countries may be more corrupt in certain types of petty corruptions and less so in the others.

For our baseline sample, we find that there is a high correlation between the overall corruption and the two petty corruption measures. The Spearman's rank correlation coefficient between overall and petty corruption (incidence and depth) equals about 0.48, significant at the 1 percent level. At the country-level, the correlations are about 0.55.

Another validity check for our corruption measures is how well they correlate with the macro-level corruption measures typically used in the literature. Thus, we check for the correlation between our overall and petty corruption measures and other corruption measures including the Control of Corruption (Worldwide Governance Indicators, World Bank), Corruption Perception Index (CPI) (Transparency International), and the corruption measure from International Country Risk Guide (ICRG). All these external corruption measures are rescaled so that higher values imply higher corruption. Doing so does not affect the analysis that follows. The correlations between the ES corruption measures (averaged at the country level) and the external ones is large and statistically significant (at the 1 percent level and 5 percent level). Table A4 in the online appendix provides the details. For instance, the rank correlation between overall corruption and the Control of Corruption variable equals 0.54, 0.44 with CPI and 0.21 with the ICRG corruption measure.

Note that the ICRG corruption measure has a relatively poorer coverage such that it is available for only 89 out of 125 countries in our sample. The Control of Corruption measure is available for all the countries and CPI is available for 118 countries.

2.4.2 *Financially constrained firms*

The ES also asks firms if they applied for a loan during the last year. If they did not, they were asked to choose the main reason for not doing so from the following list: No need for a loan as the firm has enough internal funds; application procedures were too complex; interest rates were not favorable; collateral requirements were too high; loan size and maturity was insufficient; or did not think it would be approved. For firms that did apply for a loan, the ES asks if the loan application was rejected, still pending, approved partially, or approved in full. Using this information, we first define a firm as financially unconstrained if it either did not apply for a loan because there was no need to do so (has enough internal funds) or if it applied for a loan and the loan amount was approved in full. The remaining firms are classified as financially constrained. They include, for example: firms that applied for a loan, but the application was rejected either fully or partially; firms that did not apply for a loan because they did not have the required collateral, etc. Formally, we define a dummy variable equal to 1 if the firm is financially constrained as defined above and 0 if otherwise.

As with corruption, we use average values of the financially constrained dummy variable where the average is taken over all other firms in the country-industry cell excluding the firm in question (*Financially Constrained*). In our baseline sample, the mean value of the *Financially Constrained* variable equals 0.47 and the standard deviation equals 0.20. Thus, about 47 percent

of the firms in the baseline sample are financially constrained and the rest are financially unconstrained.

2.5 *Controls*

As mentioned above, our focus is on how the relationship between corruption and employment growth varies with the financial constraints faced by the firms. Thus, while it is plausible that omitted variables may lead to spurious correlation between corruption and growth, it is much less likely that such correlation also varies with the financial constraints faced by the firm in the predicted manner. Nevertheless, the possibility cannot be ruled out completely. Thus, we control for several potential determinants of employment growth.

In the baseline specification, we control for several firm-level variables but not their interaction terms with the corruption and financial constraints variables. The interaction terms are included as controls in the robustness section. That section also includes interaction terms between some macro-level variables and corruption and financial constraints variables.

Several country-wide factors have been found to affect the overall development and growth of the private sector. Thus, their impact on job creation cannot be ruled out. Some of these factors include the overall quality of institutions for protecting property rights and maintaining the rule of law, regulatory environment, macroeconomic stability and inflation, quality of physical infrastructure availability, historical factors (ethnic fractionalization, legal origin, past colonial history, and so on), geographical factors (country size, nearness to ports), degree of trade openness, human capital availability, etc. We control for the economy-wide impact of all such factors on employment growth by using country dummies of country fixed effects. Similarly, there may be industry-wide differences in the growth of employment as sunrise industries offer greater potential

to grow than the sunset industries. Therefore, we control for such industry-wide factors using dummy variables for the industry to which the firm belongs (industry fixed effects). There are 7 industries in our sample.

Our next set of controls focuses on some important firm characteristics that are known to influence the firms' growth potential. Studies have shown that the initial firm-size and age of the firm matter in that smaller firms and younger firms tend to grow much faster than the rest (Ayyagari et al. 2011, Haltiwanger et al. 2010). Firm performance has also been linked with exporting activity, although it is not certain if exporting causes improvement in firm performance or the better performing firms self-select themselves into exporting (see for example, Bernard and Jensen 1999, Melitz 2003, Wagner 2007). Similarly, firms with foreign ownership have also been found to enjoy numerous benefits such as better access to modern technology, greater access to international markets, among others. These benefits are likely to impact firms' growth rate. Thus, we control for (log of) number of workers at the firm at the beginning of the growth accounting period (*Initial Employment*); (log of) age of the firm; a dummy variable equal to 1 if the firm is part of a larger organization and 0 otherwise *(Multi-firm*) as a proxy for size related benefits; the proportion of firms' annual sales made abroad *(Exports*) and the proportion of firms' ownership that is with foreign individuals and companies (*Foreign Ownership*).

Several studies have analyzed gaps in firm performance stemming from the manager's gender and quality. Firms owned or managed by women tend to perform worse than those owned or managed by men (Coleman 2000, Du Rietz and Henrekson 2000, Sabarwal and Terrell 2008). This is likely to be due to differences in the size of the firms managed by women, the difficulty that women face relative to men in obtaining finance, and social attitudes that tend to discourage women's economic participation. Thus, we account for the gender of the top manager with a

dummy variable indicating if the top manager of the firm is a female or not (*Female Top Manager*). Differences in the quality of management, for reasons other than the gender of the top manager, have been found to impact firm productivity and performance (see for example, Bloom and Van Reenen 2007, Syverson 2011, Pfeifer 2015). Differences in education and experience of the top manager could be the possible factors driving differences in management quality. We account for this in our regressions using a proxy measure which equals (log of) the number of years of experience the top manager of the firm has working in the industry (*Top Manager Experience*).

Recent investments in capital stock to expand the productive capacity of the firm are likely to impact growth rate positively. Information in the ES is available on capital stock (replace and book value of machines and equipment). However, due to the non-response, we use an alternative proxy measure of investment. This is a dummy variable equal to 1 if the firm purchased fixed assets during the last year and 0 if otherwise. This variable has also been used as a proxy for capital use in Amin and Soh (2020) and Islam et al. (2019).

Our next set of controls includes several aspects of the business environment. Above, we mentioned that the control for country dummies or country fixed effects accounts for all economy-wide aspects of the economies including the business environment. However, the country fixed effects do not account for differences in the business environment across firms within a country. Such differences may arise because of differences in the implementation of rules across regions within a country, regional disparities in economic development, differences in firm capacities to deal with factors such as power outages, access to finance, etc. Thus, we control for the following firm-level variables: (log of 1 plus) total hours of power outage in a typical month over the last year (*Power Outage*); dummy variable equal to 1 if the firm competes against informal firms and 0 otherwise (*Informal Competition*); severity level of courts functioning (0-4 scale) as an obstacle

for firms' current operations (*Courts: Obstacle Severity*); percentage of senior management's time spent in dealing with business regulations (*Time Tax*); losses due to crime as a percentage of firms' annual sales (*Crime Losses*); severity level of tax rates (0-4 scale) as an obstacle for firms' current operations (*Taxes: Obstacle Severity*); and severity level of labor laws (0-4 scale) as an obstacle for firms' current operations (*Labor Laws: Obstacle Severity*).

3. Base regression results

3.1 *Overall Corruption*

Our baseline regression results for overall corruption are provided in Table 1. As mentioned above, country and industry fixed effects are used in all the specifications. The remaining controls are added sequentially.

For all the specifications in Table 1, the estimated coefficient value of the interaction term between corruption and financial constraints is negative, large and statistically significant at the 1 percent level. Thus, corruption hurts more (or benefits less) when firms are financially constrained compared to when they are not financially constrained. Across the different specifications, the estimated coefficient value of the interaction term ranges narrowly between -1.26 in the final specification (column 4) and -1.37 (column 1). Thus, based on our final specification with all the controls included and our most conservative estimate, one percentage point increase in overall corruption or bribe rate is associated with a decrease in employment growth by 1.26 percentage points more for a firm that faces the highest level of financial constraints than for a firm that faces the least level of financial constraints. This is a large difference given that mean level of employment growth in our sample is 5.05 percent.

For the total impact of overall corruption, consider the final specification in Table 1 (column 4). A one percentage point increase in overall corruption is associated with a decrease in employment growth rate by 0.48 percentage points for the most financially constrained firm. For the least financially constrained firm, the unit increase in overall corruption is associated with an increase in employment growth rate by 0.78 percentage points. Both these changes in employment growth rate are significant at the 5 percent level.

Regarding the various controls, both firm-size and age of the firm are significantly inversely correlated (at the 1 percent level) with employment growth. Employment growth is also significantly lower (at the 1 percent level) for firms with a female top manager vs. those with a male top manager; and significantly higher at the 5 percent level or less for firms that have foreign ownership, are part of a bigger firm (*Multi-firm*), and those that bought fixed assets over the last two years (significant at the 1 percent level). Last, firms that face informal competition show lower employment growth than firms that do not face such competition. However, this relationship is significant only at the 10 percent level and hence should be treated with due caution.

3.2 *Petty Corruption*

Regression results for petty corruption (incidence and depth) are provided in Table 2. For brevity, only some of the specifications are shown. Results for the full set of specifications are provided in Tables A5 and A6 in the online appendix. Results for petty corruption are consistent with the ones above for overall corruption. That is, the interaction term between petty corruption and financial constraints is negative, large and statistically significant at the 5 percent level in one specification and 1 percent level in all the other specifications.

Quantitatively, based on the final specifications in Table 2 with all the controls included, a 10 percentage points increase in the incidence of petty corruption is associated with a decrease in employment growth rate by 0.67 percentage point (significant at the 10 percent level) for the most financially constrained firms, and an increase in employment growth rate by 1.06 percentage points (significant at the 1 percent level) for the least financially constrained firms. Similarly, for each 10 percentage points increase in the depth of petty corruption, employment growth rate falls by 0.98 percentage point (significant at the 5 percent level) for the financially most constrained firms and it increases by 1.29 percentage points (significant at the 1 percent level) for the least financially constrained firms. These are economically large differences in how corruption impacts the least and most financially constrained firms.

Results for the various controls are like the ones discussed above in the case of overall corruption. The only change is that employment growth is higher for firms that report higher power outages. However, the result should be treated with due caution as it is significant only at the 10 percent level and we found no such significant relationship when using overall corruption.

4. Robustness

4.1 *Controlling for other interaction terms*

Our first robustness check involves controlling for several non-linear effects (interaction terms and squared terms). These controls include the following: interaction terms between corruption and several firm-level controls used in the baseline specification above; interaction terms between financial constraints variable and firm-level controls used above; square of corruption and financial constraints variables; interaction term between overall economic development (GDP per capita) and its growth rate with corruption and financial constraints.

Regression results with the interaction terms mentioned in the previous paragraph along with all the other controls used in the baseline model (Section 3) are provided in Table 3. Columns 1 and 2 contain results for overall corruption; columns 3 and 4 contain results for the incidence of petty corruption while columns 5 and 6 contain results for the depth of petty corruption. For brevity, results are shown for only some of the specifications. Results for other specifications are provided in Table A7 through A9 in the online appendix.

These results show that the interaction term between corruption and financial constraints remains negative, large and statistically significant at the 5 percent level for overall corruption (columns 1 and 2) and at the 1 percent level for incidence and depth of petty corruption (columns 3 through 6). There is not much change in the value of the interaction term between overall corruption and financial constraints from the final baseline model. For both measures of petty corruption, the interaction term is slightly bigger (more negative) than in the baseline model. For instance, the estimated coefficient value of the interaction term between depth of petty corruption and financial constraints equals -0.23 in the final baseline specification (column 4, Table 2); it increases (in absolute value) to -0.26 when we add the various other interaction term controls (column 6, Table 3).

Regarding the various other interaction terms included as controls, the negative relationship between corruption and employment growth is enhanced for firms that face higher power outages than for firms that face lower power outages. The difference is significant at the 5 percent level for overall corruption and 1 percent level for the two petty corruption measures. Thus, the implication here is that lack of public amenities as reflected in high levels of power outages makes corruption even more harmful. We also find that the negative relationship between petty corruption (incidence and depth) and employment growth is much stronger (more negative) at low

levels of corruption than at high levels of corruption. This difference in corruption's impact is significant at the 1 percent level. Thus, initial levels of petty corruption hurt employment growth more. However, overall corruption shows no such pattern. Next, the impact of financial constraints on employment growth is significantly less adverse in countries with higher growth rate of GDP per capita. The difference is significant at the 1 percent level. Some other interaction terms are also significant but not consistently so across all the corruption indicators.

4.2 *Alternative measure of financial constraints*

As discussed above, we check for the robustness of our financial constraint variable by using the proportion of firms in the country-industry cell (excluding the firm in question) that report access to finance as a major obstacle to their current operations. Regressions results using the baseline specification are provided in the online appendix in Table A10 (for overall corruption) and Table A11 (for both petty corruption measures). Our main result continues to hold. That is, the interaction term between corruption and the financial obstacle variable is large, negative and statistically significant at 1 percent level for overall corruption and 5 percent level for both measures of petty corruption. Thus, corruption hurts employment growth much more for firms that find access to finance to be major obstacle for their current operations than the rest.

4.3 *Alternative measure of employment growth*

Next, we check our baseline results for overall and petty corruption using log differences in employment as the measure of employment growth (*Employment Growth* 1). Regression results are provided in Table A12 and A13 in the online appendix for overall corruption and petty corruption, respectively. The results show that the interaction term between corruption and

financial constraints remains large, negative and statistically significant at the 5 percent level in one specification and 1 percent level in the remaining specifications.

4.4 *Corruption-finance nexus and institutional quality*

The premise for the results above is that having better access to finance can relieve some of the problems due to corruption. That is, firms can pay their way through corruption provided they have the money to do so. As discussed above, this is more likely when there is weak implementation of the laws, allowing public officials to favor those who can pay the bribe. Thus, if our results are indeed picking up the effect of corruption and finance nexus on employment growth then we can expect them to be stronger when law enforcement is weaker. However, if our results are spuriously driven, there is no reason to expect the corruption-finance relationship to vary with the quality of law enforcement.

For the quality of law enforcement, we use the question in the ES on the severity of the lack of proper functioning courts as an obstacle to firms' operations. As with corruption and financial constraint variables, we use CI cell averages of the responses of firms to the question on courts functioning, where the average is taken over all firms in the cell excluding the firm in question (*Courts: Obstacle Severity*). The null hypothesis is that the triple interaction term between corruption, financial constraint and the severity level of courts as an obstacle is negative, implying that the corruption-finance interaction term which we found to be negative above becomes even more negative when the lack of proper functioning of the courts is a more serious obstacle to the firms.

Regression results are provided in Table 4. For brevity, only some of the specifications are shown. Results for the full set of specifications are provided in Tables A14 and A15 in the online

appendix. These results confirm the prediction in the previous paragraph. That is, the triple interaction term is negative, large and statistically significant at the 5 percent level. This holds when using the overall corruption measure as well as the two petty corruption measures. This helps raise our confidence against endogeneity problems with our main results.

4.5 *Falsification test*

The next check against endogeneity is based on the idea that while having money to pay bribes can result in a quid pro quo situation alleviating some of the negative effects of corruption, the quid pro quo is much less or non-existent with other elements of the business environment correlated with corruption. Thus, if the corruption-finance nexus found above is indeed about corruption then we should not find any significant difference in how other correlated elements of the business environment affect firms that face high versus low financial constraints. However, if our corruption measure is a mere proxy for other correlated elements of the business environment, it should be the case that the correlated elements of the business environment impact high and low financially constrained firms differently like we found for corruption.

We focus on three elements of the business environment which include physical infrastructure availability, regulatory burden experienced by the firms, and the quality of enforcement of the laws. We proxy these by CI cell averages (for all firms excluding the firm in question) of total hours of power outage (*Power Outage*), percentage of senior management's time spent in dealing with business regulations (*Time Tax*), and severity of the lack of proper functioning courts as an obstacle to firms' operations (*Courts: Obstacle Severity*). All these variables have been discussed above. The regression exercise involves repeating the baseline regression (as in Table 1) but substituting the corruption and financial constraint interaction term

with the interaction terms between financial constraints (cell averages as above) and the cell averages of: (i) power outages; (ii) time tax; and (iii) courts quality, all in separate regressions.

Regression results are provided in Table 5. Columns 1 and 2 contain results for power outages, columns 3 and 4 contain results for time tax, and columns 5 and 6 contain the results for courts quality. For brevity, only some of the baseline specifications are shown. As evident from Table 5, none of the interaction terms is statistically significant. That is, unlike corruption, being financially constrained or not has no implications for how the infrastructure availability, regulatory burden and courts quality impacts employment growth rate. Thus, it is highly unlikely that our results above for the corruption-finance nexus are capturing the interplay between access to finance and the broader elements of the business environment.

5. Conclusion

The payment of bribes and expenses incurred on rent-seeking activity associated with corruption can make a significant demand on firms' financial resources. Thus, firms that do not have enough cash of their own and face high cost of borrowing money are likely to find corruption more burdensome than firms that have easy access to finance. This heterogeneity in the effects of corruption raises the possibility of reconciling the hitherto two disparate and mutually exclusive models of corruption, the "sands the wheels" and "grease the wheels" models. That is, corruption can hurt some firms and at the same time benefit others depending on their underlying conditions such as access to finance.

The present paper applies the above idea to the impact of corruption on employment growth for a large cross-section of private firms. The impact of corruption on employment growth or job creation has not been discussed in the literature, let alone how the effect is mediated through access

to finance. Our empirical results confirm that corruption has a much larger negative effect on employment growth for firms that are relatively more financially constrained. As suggested in the previous paragraph, we find evidence that corruption helps "grease the wheels" for firms that have easy access to finance but "sands the wheels" for firms that face high levels of financial constraints.

Several issues remain to be explored. We provide a few examples to illustrate the point. First, it is important to check if our results extend to other aspects of firm performance such as labor and total factor productivity, sales growth, investment in physical capital, and innovation. Second, factors other than access to finance may magnify or mitigate the effects of corruption. For instance, higher corruption tilts the cost advantage away from formal and towards informal firms as the latter can escape much of corruption due to their unofficial status. Thus, formal firms that compete against informal firms may suffer more from corruption that formal firms that do not compete against the informal firms. Third, our analysis focused on the private firms. Corruption affects not just the private firms but also the households in availing various government services. It will be interesting to check if financial constraints play a role in how corruption affects individuals and households. Last, it will be useful to explore the role of access to finance in determining the level of corruption itself. Some studies show that corrupt public officials tend to target the more profitable and faster growing firms as these firms have greater ability to pay bribes. The logic can be extended to finance as better access to finance raises the firms' ability to pay bribes. We hope that the present paper motivates future research along these and other related lines.

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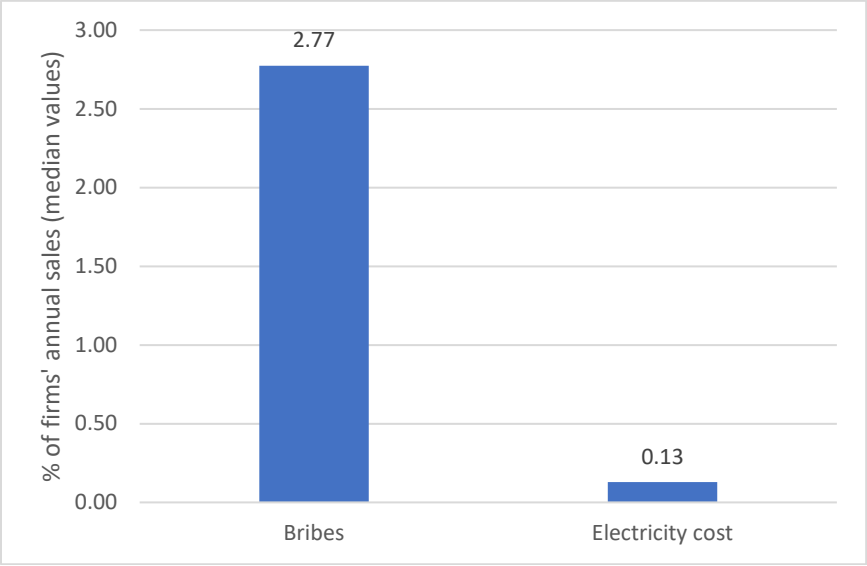
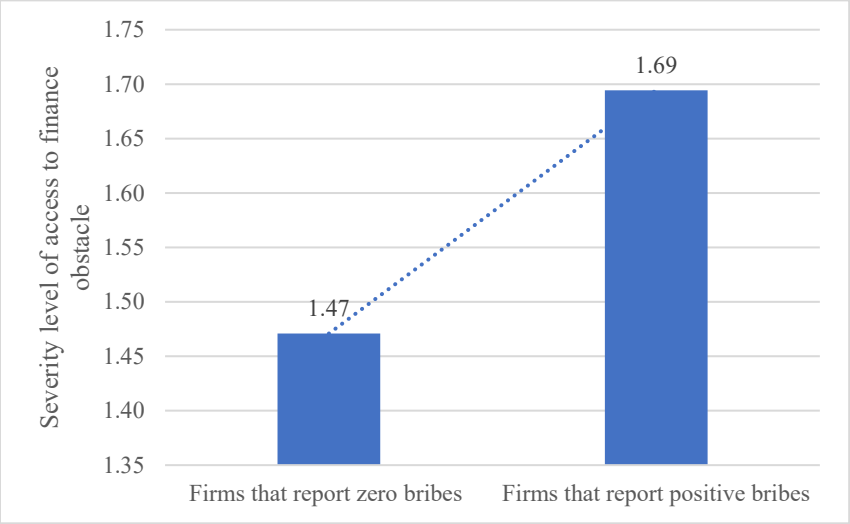
Figure 1: Reported bribes vs. other costs (% of firms' annual sales)



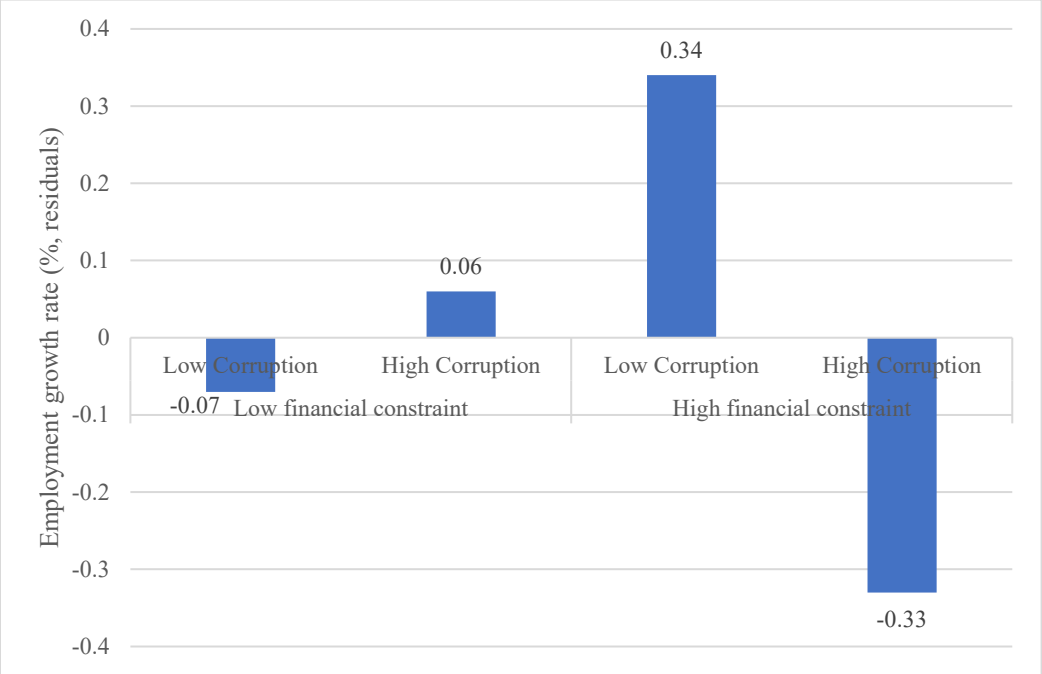
Source: Authors' own calculations based on Enterprise Surveys data.
Note: The sample is restricted to only those firms that report positive bribes paid to public officials "to get things done". It excludes firms with missing data on electricity cost of bribe amount paid. The figure uses sampling weights provided in the ES. The qualitative nature of the results shown in the figure remains unchanged even without the use of sampling weights.

Figure 2: Bribe payment and the severity level of access to finance as an obstacle to firms' current operations



Source: Authors' own calculations based on Enterprise Surveys data.
Note: The bribe variable used here is based on the question in the ES that asked firms about the amount of bribes firms like itself pay to public officials to "get things done". The figure uses sampling weights provided in the ES. The qualitative nature of the results shown in the figure remains unchanged even without the use of sampling weights. The severity level of access to finance (Y-axis variable) is measured on 0-4 scale as No obstacle (0), minor obstacle (1), moderate obstacle (2), major obstacle (3) and a very severe obstacle (4).

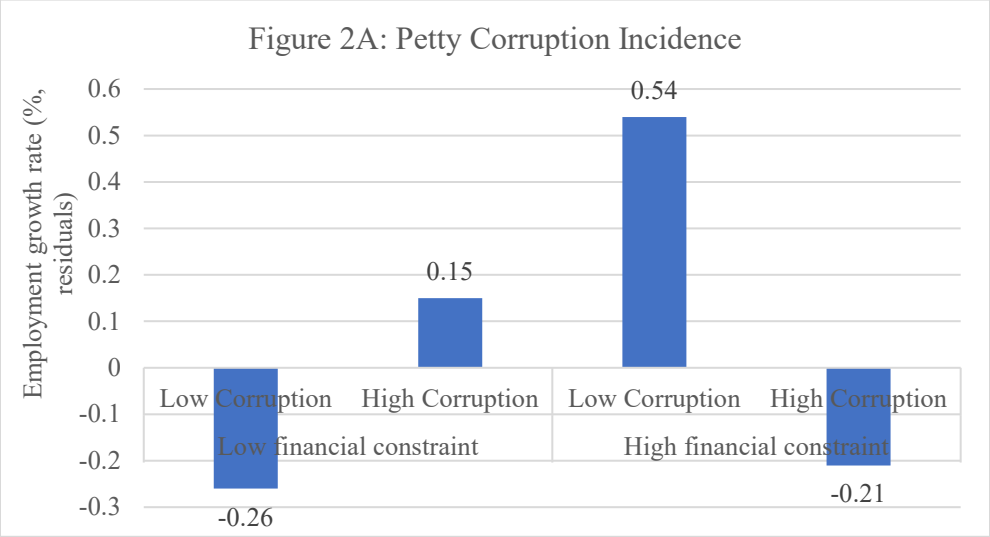
Figure 3: Impact of increase in overall corruption on employment growth for firms that face low vs. high levels of financial constraints

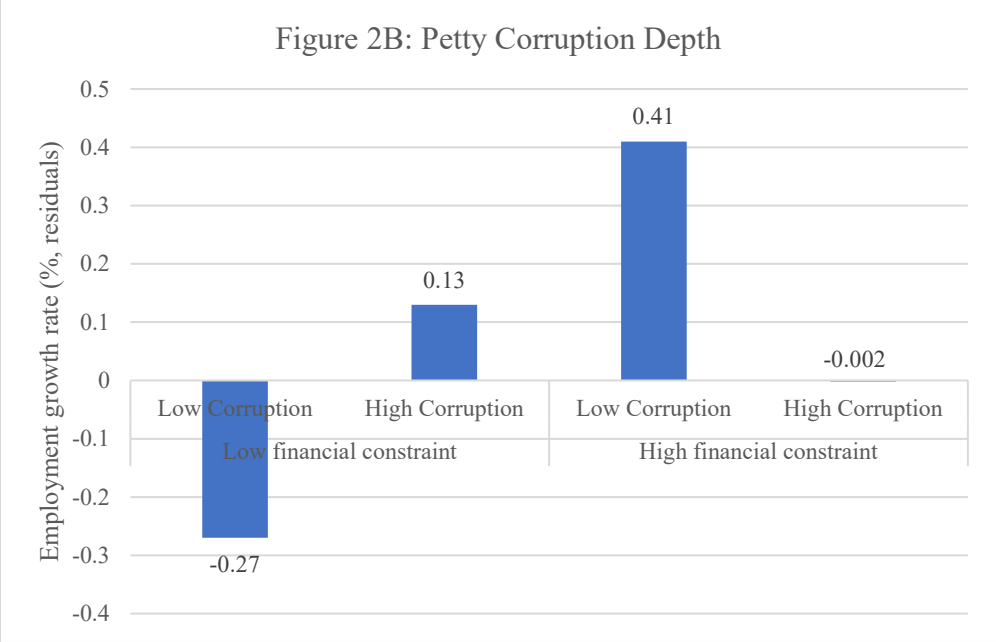


Source: Enterprise Surveys. Based on authors' own calculations.

Notes: The figure is a partial scatter plot of the residuals of growth rate of employment plotted against residuals of overall corruption and residuals of financial constraints variable. The residuals of employment growth are obtained by regression employment growth on country fixed effects, industry fixed effects and initial employment at the firm. Residuals of overall corruption are obtained by regressing overall corruption on country fixed effects, industry fixed effects, initial employment at the firm and financial constraints variable. Residuals of financial constraints variable are obtained like for overall corruption with the control for financial constraints replaced by overall corruption variable. The cut-off used for low vs. high financial constraints groups in the graph is the median value of the residuals of financial constraints variable. For overall corruption, the cut-off used is its median value within each of the two financial constraints groups shown. All variables are as defined in the main text.

Figure 4: Impact of increase in petty corruption on employment growth for firms that face low vs. high levels of financial constraints





Source: Enterprise Surveys. Based on authors' own calculations.

Notes: The figure is a partial scatter plot of the residuals of growth rate of employment plotted against residuals of petty corruption (Petty Corruption Incidence in Figure 4A and Petty Corruption Depth in Figure 4B) and residuals of financial constraints variable. The residuals of employment growth are obtained by regression employment growth on country fixed effects, industry fixed effects and initial employment at the firm. Residuals of petty corruption are obtained by regressing petty corruption on country fixed effects, industry fixed effects, initial employment at the firm and financial constraints variable. Residuals of financial constraints variable are obtained like for petty corruption with the control for financial constraints replaced by petty corruption variable. The cut-off used for low vs. high financial constraints groups in the graph is the 75th percentile value of the residuals of financial constraints variable. For petty corruption, the cut-off used is its median value within each of the two financial constraints groups shown. All variables are as defined in the main text.

Table 1: Base Regression Results, OLS				
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**
Overall Corruption (CI avg.) * Financially Constrained (CI avg.)	-1.374***	-1.363***	-1.268***	-1.260***
	(0.460)	(0.440)	(0.445)	(0.445)
Overall Corruption (CI avg.)	0.836**	0.820***	0.783**	0.775**
	(0.329)	(0.317)	(0.321)	(0.321)
Financially Constrained (CI avg.)	1.228	1.380	1.338	1.344
	(1.708)	(1.751)	(1.758)	(1.765)
Initial Employment (logs)	-5.760***	-5.608***	-5.872***	-5.874***
	(0.348)	(0.325)	(0.322)	(0.328)
Age of Firm (logs)		-2.138***	-2.033***	-2.034***
		(0.296)	(0.294)	(0.294)
Manager Experience (logs)		0.103	0.011	0.022
		(0.273)	(0.275)	(0.274)
Exports (proportion of sales)		1.324	1.095	1.099
		(2.157)	(2.189)	(2.198)
Foreign Ownership (% ownership)		0.024***	0.022**	0.022**
		(0.009)	(0.009)	(0.009)
Multi-firm (dummy)		1.819**	1.745**	1.733**
		(0.705)	(0.715)	(0.709)
Female Manager (dummy)			-1.230***	-1.230***
			(0.444)	(0.443)
Power Outage (logs)			0.193	0.200
			(0.124)	(0.122)
Informal Competition (dummy)			-0.859*	-0.832*
			(0.448)	(0.435)
Firm Bought Assets (dummy)			3.022***	3.043***
			(0.440)	(0.455)
Time Tax (%)				0.004
				(0.015)
Courts: Obstacle Severity				0.012
				(0.213)
Crime Losses (% of sales)				-0.014
				(0.046)
Taxes: Obstacle Severity				-0.161
				(0.166)
Labor Laws: Obstacle Severity				0.002
				(0.174)
Industry fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes
Number of observations	41,203	41,203	41,203	41,203
R-squared	0.167	0.177	0.186	0.186

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table 2: Base regression results for petty corruption

	Petty Corruption Incidence		Petty Corruption Depth	
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**
Petty Corruption Incidence (CI avg.) * Financially Constrained (CI avg.)	-0.168**	-0.173***		
	(0.066)	(0.065)		
Petty Corruption Incidence (CI avg.)	0.097**	0.106***		
	(0.038)	(0.037)		
Petty Corruption Depth (CI avg.) * Financially Constrained (CI avg.)			-0.218***	-0.228***
			(0.079)	(0.078)
Petty Corruption Depth (CI avg.)			0.117***	0.129***
			(0.044)	(0.044)
Financially Constrained (CI avg.)	3.004	3.546*	3.076	3.674*
	(1.969)	(2.043)	(1.928)	(1.994)
Initial Employment (logs)	-5.763***	-5.889***	-5.764***	-5.888***
	(0.322)	(0.314)	(0.322)	(0.314)
Age of Firm (logs)		-1.950***		-1.954***
		(0.291)		(0.291)
Manager Experience (logs)		-0.068		-0.072
		(0.274)		(0.274)
Exports (proportion of sales)		1.406		1.399
		(1.842)		(1.840)
Foreign Ownership (% ownership)		0.021**		0.021**
		(0.008)		(0.008)
Multi-firm (dummy)		1.833***		1.828***
		(0.645)		(0.645)
Female Manager (dummy)		-1.334***		-1.327***
		(0.435)		(0.435)
Power Outage (logs)		0.211*		0.213*
		(0.120)		(0.120)
Informal Competition (dummy)		-0.725*		-0.727*
		(0.421)		(0.421)
Firm Bought Assets (dummy)		3.135***		3.134***
		(0.423)		(0.423)
Time Tax (%)		0.005		0.005
		(0.013)		(0.013)
Courts: Obstacle Severity		-0.033		-0.033
		(0.198)		(0.198)
Crime Losses (% of sales)		-0.024		-0.025
		(0.045)		(0.045)
Taxes: Obstacle Severity		-0.172		-0.171
		(0.164)		(0.164)

Labor Laws: Obstacle Severity		-0.004		-0.003
		(0.171)		(0.171)
Industry fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes
Number of observations	41,117	41,117	41,117	41,117
R-squared	0.167	0.186	0.167	0.186

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table 3: Robustness for other interaction terms						
	Overall Corruption		**Petty Corruption Incidence**		**Petty Corruption Depth**	
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Overall Corruption (CI avg.) * Financially Constrained (CI avg.)	**-1.160****	**-1.206****				
	(0.478)	(0.503)				
Overall Corruption (CI avg.)	1.008***	1.974				
	(0.322)	(1.395)				
Petty Corruption Incidence (CI avg.) * Financially Constrained (CI avg.)			**-0.191*****	**-0.210*****		
			(0.061)	(0.071)		
Petty Corruption Incidence (CI avg.)			-0.020	-0.456**		
			(0.052)	(0.201)		
Petty Corruption Depth (CI avg.) * Financially Constrained (CI avg.)					**-0.251*****	**-0.256*****
					(0.074)	(0.085)
Petty Corruption Depth (CI avg.)					0.012	-0.365
					(0.058)	(0.229)
Financially Constrained (CI avg.)	-2.605	-27.505	1.772	-3.266	2.169	-5.046
	(5.018)	(18.112)	(4.810)	(16.698)	(4.807)	(16.367)
Initial Employment (logs)	-5.876***	-5.510***	-5.883***	-6.226***	-5.878***	-6.196***
	(0.328)	(0.776)	(0.313)	(0.811)	(0.312)	(0.803)
Age of Firm (logs)	-2.030***	-1.822***	-1.959***	-1.593**	-1.964***	-1.662**
	(0.294)	(0.685)	(0.291)	(0.708)	(0.292)	(0.700)
Manager Experience (logs)	0.025	0.029	-0.082	-0.050	-0.089	-0.059
	(0.274)	(0.274)	(0.273)	(0.271)	(0.274)	(0.272)
Exports (proportion of sales)	1.072	4.524*	1.320	5.361**	1.364	5.639**
	(2.198)	(2.548)	(1.833)	(2.428)	(1.831)	(2.619)
Foreign Ownership (% ownership)	0.022**	-0.015	0.021**	-0.011	0.021**	-0.012
	(0.009)	(0.018)	(0.008)	(0.019)	(0.008)	(0.019)
Multi-firm (dummy)	1.741**	1.902***	1.853***	1.991***	1.853***	1.993***
	(0.709)	(0.685)	(0.643)	(0.611)	(0.642)	(0.605)
Female Manager (dummy)	-1.227***	-1.218***	-1.376***	-1.378***	-1.379***	-1.389***
	(0.443)	(0.442)	(0.432)	(0.432)	(0.432)	(0.429)
Power Outage (logs)	0.201	0.253	0.218*	0.500*	0.220*	0.446
	(0.122)	(0.263)	(0.120)	(0.279)	(0.121)	(0.275)
Informal Competition (dummy)	-0.846*	-0.637	-0.728*	-1.107	-0.714*	-1.131
	(0.435)	(0.990)	(0.421)	(1.033)	(0.422)	(1.028)
Firm Bought Assets (dummy)	3.047***	3.964***	3.125***	4.404***	3.126***	4.414***
	(0.455)	(0.953)	(0.423)	(0.942)	(0.422)	(0.935)
Time Tax (%)	0.004	-0.008	0.006	-0.002	0.005	-0.006

	(0.014)	(0.028)	(0.013)	(0.029)	(0.013)	(0.029)
Courts: Obstacle Severity	0.011	0.052	-0.041	-0.214	-0.037	-0.196
	(0.213)	(0.401)	(0.199)	(0.401)	(0.199)	(0.400)
Crime Losses (% of sales)	-0.013	0.080	-0.022	0.064	-0.023	0.070
	(0.046)	(0.106)	(0.045)	(0.106)	(0.046)	(0.105)
Taxes: Obstacle Severity	-0.160	-0.168	-0.174	-0.190	-0.176	-0.192
	(0.166)	(0.166)	(0.163)	(0.166)	(0.163)	(0.167)
Labor Laws: Obstacle Severity	0.004	0.003	0.003	0.005	0.000	0.002
	(0.174)	(0.174)	(0.170)	(0.169)	(0.171)	(0.169)
Square of Financially Constrained (CI avg.)	3.844	2.575	2.169	-0.327	1.858	-0.497
	(4.863)	(4.763)	(4.659)	(4.657)	(4.652)	(4.644)
Financially Constrained (CI avg.) * Initial Employment		-0.407		-0.128		0.031
		(1.421)		(1.376)		(1.379)
Financially Constrained (CI avg.) * Age of Firm		-0.703		-0.525		-0.574
		(1.311)		(1.301)		(1.303)
Financially Constrained (CI avg.) * Exports		-7.446		-7.409		-7.773
		(6.168)		(5.455)		(5.485)
Financially Constrained (CI avg.) * Foreign Ownership		0.072**		0.078**		0.076**
		(0.035)		(0.033)		(0.033)
Financially Constrained (CI avg.) * Firm Bought Assets		-1.424		-1.185		-1.465
		(1.765)		(1.752)		(1.745)
Financially Constrained (CI avg.) * Power Outage		0.242		0.423		0.386
		(0.508)		(0.472)		(0.478)
Financially Constrained (CI avg.) * Time Tax		0.014		0.024		0.023
		(0.057)		(0.057)		(0.057)
Financially Constrained (CI avg.) * Courts: Obstacle Severity		-0.264		-0.714		-0.749
		(0.752)		(0.767)		(0.770)
Financially Constrained (CI avg.) * Informal Competition		0.147		-0.709		-0.564
		(1.844)		(1.834)		(1.822)
Financially Constrained (CI avg.) * Crime Losses		-0.208		-0.253		-0.240
		(0.220)		(0.193)		(0.187)
Financially Constrained (CI avg.) * GDP per capita (logs)		2.860		0.776		0.995
		(1.781)		(1.623)		(1.596)
Financially Constrained (CI avg.) * Growth rate of GDP per capita		1.795***		1.503***		1.461***
		(0.545)		(0.477)		(0.475)
Corruption in the rows below is Overall Corruption (columns 1-2), Petty Corruption Incidence (columns 3-4) and Petty Corruption Depth (columns 5-6)						
Square of Corruption (CI avg.)	-0.021	-0.016	0.002***	0.003***	0.002***	0.003***
	(0.017)	(0.014)	(0.001)	(0.001)	(0.001)	(0.001)
Corruption (CI avg.) * Initial Employment		-0.189		0.022		0.020
		(0.157)		(0.019)		(0.023)
Corruption (CI avg.) * Age of Firm		0.118		-0.006		-0.001
		(0.143)		(0.015)		(0.017)

Corruption (CI avg.) * Exports		-0.207		-0.048		-0.064
		(0.679)		(0.104)		(0.145)
Corruption (CI avg.) * Foreign ownership		0.002		-0.000		-0.000
		(0.006)		(0.000)		(0.000)
Corruption (CI avg.) * Firm Bought Assets		-0.253		-0.045*		-0.047
		(0.164)		(0.024)		(0.029)
Corruption (CI avg.) * Power Outage		-0.100**		-0.023***		-0.024***
		(0.045)		(0.007)		(0.008)
Corruption (CI avg.) * Time Tax		0.004		-0.000		-0.000
		(0.006)		(0.001)		(0.001)
Corruption (CI avg.) * Courts: Obstacle severity		0.066		0.030**		0.038***
		(0.074)		(0.011)		(0.015)
Corruption (CI avg.) * Informal competition		-0.216		0.038*		0.047*
		(0.173)		(0.022)		(0.026)
Corruption (CI avg.) * Crime Losses		0.004		0.002		0.001
		(0.012)		(0.002)		(0.003)
Corruption (CI avg.) * GDP per capita (logs)		-0.067		0.045**		0.036
		(0.132)		(0.021)		(0.024)
Corruption (CI avg.) * Growth rate of GDP per capita		0.006		-0.001		-0.000
		(0.015)		(0.004)		(0.005)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	41,203	41,108	41,117	41,022	41,117	41,022
R-squared	0.186	0.190	0.187	0.193	0.187	0.192

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%).

Table 4: Corruption-Finance-Institutional Quality nexus

	Overall Corruption		Petty Corruption Incidence		Petty Corruption Depth	
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Overall Corruption (CI avg.) * Financially Constrained (CI avg.) * Courts: Obstacle Severity (CI avg.)	**-1.526*** ** (0.553)	**-1.507*** ** (0.539)				
Petty Corruption Incidence (CI avg.) * Financially Constrained (CI avg.) * Courts: Obstacle Severity (CI avg.)			**-0.237** ** (0.103)	**-0.186* ** (0.102)		
Petty Corruption Depth (CI avg.) * Financially Constrained (CI avg.) * Courts: Obstacle Severity (CI avg.)					**-0.328*** ** (0.119)	**-0.270** ** (0.116)
Overall Corruption (CI avg.) * Financially Constrained (CI avg.)	0.543 (0.777)	0.633 (0.787)				
Overall Corruption (CI avg.) * Courts: Obstacle Severity (CI avg.)	1.021*** (0.343)	1.004*** (0.340)				
Overall Corruption (CI avg.)	-0.281 (0.414)	-0.323 (0.426)				
Petty Corruption Incidence (CI avg.) * Financially Constrained (CI avg.)			0.131 (0.134)	0.062 (0.132)		
Petty Corruption Incidence (CI avg.) * Courts: Obstacle Severity (CI avg.)			0.096* (0.053)	0.065 (0.052)		
Petty Corruption Incidence (CI avg.)			-0.030 (0.074)	0.017 (0.074)		

Petty Corruption Depth (CI avg.) * Financially Constrained (CI avg.)					0.176	0.094
					(0.156)	(0.155)
Petty Corruption Depth (CI avg.) * Courts: Obstacle Severity (CI avg.)					0.146**	0.109*
					(0.060)	(0.059)
Petty Corruption Depth (CI avg.)					-0.068	-0.011
					(0.086)	(0.087)
Financially Constrained (CI avg.) * Courts: Obstacle Severity (CI avg.)	-0.801	-0.469	0.968	0.880	1.115	1.128
	(2.461)	(2.456)	(2.932)	(2.939)	(2.845)	(2.843)
Financially Constrained (CI avg.)	2.097	1.964	1.756	2.522	1.804	2.505
	(2.752)	(2.837)	(3.373)	(3.465)	(3.300)	(3.396)
Courts: Obstacle Severity (CI avg.)	-0.255	-0.731	-0.535	-0.736	-0.746	-0.977
	(1.435)	(1.455)	(1.549)	(1.576)	(1.512)	(1.535)
Initial Employment (logs)	-5.754***	-5.867***	-5.758***	-5.883***	-5.764***	-5.886***
	(0.348)	(0.328)	(0.321)	(0.313)	(0.320)	(0.312)
Age of firm (logs)		-2.041***		-1.957***		-1.957***
		(0.293)		(0.291)		(0.291)
Manager Experience (logs)		0.040		-0.060		-0.062
		(0.274)		(0.273)		(0.273)
Exports (proportion of sales)		1.071		1.410		1.397
		(2.198)		(1.822)		(1.817)
Foreign Ownership (% ownership)		0.022**		0.021**		0.021**
		(0.009)		(0.008)		(0.008)
Multi-firm (dummy)		1.756**		1.818***		1.806***
		(0.712)		(0.645)		(0.646)
Female Manager (dummy)		-1.229***		-1.343***		-1.333***
		(0.443)		(0.434)		(0.433)
Power Outage (logs)		0.188		0.203*		0.204*
		(0.123)		(0.120)		(0.120)
Informal competition (dummy)		-0.841*		-0.706*		-0.714*
		(0.440)		(0.425)		(0.424)
Firm Bought Assets (dummy)		3.053***		3.122***		3.121***
		(0.454)		(0.421)		(0.420)
Time Tax (%)		0.005		0.006		0.006
		(0.015)		(0.013)		(0.013)
Crime Losses (% of sales)		-0.011		-0.023		-0.024
		(0.046)		(0.045)		(0.045)
Taxes: Obstacle Severity		-0.161		-0.176		-0.175

		(0.172)		(0.165)		(0.165)
Labor Laws: Obstacle		0.009		-0.008		-0.005
Severity		(0.166)		(0.165)		(0.165)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Constant	19.659***	23.830***	19.410***	22.998***	19.657***	23.251***
	(2.329)	(2.385)	(2.560)	(2.632)	(2.537)	(2.605)
Number of observations	41,197	41,197	41,108	41,108	41,108	41,108
R-squared	0.168	0.187	0.168	0.187	0.168	0.187

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%).

Table 5: Falsification test

	Power Outages		Time Tax (Regulatory burden)		Courts Obstacle (severity level)	
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Financially Constrained (CI avg.) * Power Outage (CI avg.)	**-0.447**	**-0.545**				
	(1.286)	**(1.281)**				
Financially Constrained (CI avg.) * Time Tax (CI avg.)			**-0.140**	**-0.129**		
			(0.162)	**(0.167)**		
Financially Constrained (CI avg.) * Courts: Obstacle Severity (CI avg.)					**-3.298**	**-2.851**
					(2.246)	**(2.258)**
Courts: Obstacle Severity (CI avg.)	1.194	0.720		-0.751		-0.701
	(1.307)	(1.332)		(0.775)		(0.770)
Power Outage (logs, CI avg.)		-0.259	0.002	0.059		-0.265
		(0.585)	(0.912)	(0.914)		(0.587)
Time Tax (%, CI avg.)		-0.024		-0.027	0.036	0.033
		(0.058)		(0.058)	(0.080)	(0.081)
Financially Constrained (CI avg.)	2.951	2.791	0.238	0.775	1.106	1.345
	(2.633)	(2.708)	(2.222)	(2.254)	(2.305)	(2.382)
Initial Employment (logs)	-5.748***	-5.873***	-5.763***	-5.872***	-5.772***	-5.866***
	(0.347)	(0.323)	(0.348)	(0.323)	(0.346)	(0.322)
Age of Firm (logs)		-2.062***		-2.065***		-2.064***
		(0.292)		(0.292)		(0.292)
Manager Experience (logs)		0.039		0.034		0.034
		(0.274)		(0.274)		(0.274)
Exports (proportion of sales)		1.085		1.086		1.080
		(2.199)		(2.200)		(2.201)
Foreign Ownership (% ownership)		0.023**		0.023**		0.023**
		(0.009)		(0.009)		(0.009)
Multi-firm (dummy)		1.689**		1.693**		1.689**
		(0.714)		(0.713)		(0.714)
Female Manager (dummy)		-1.229***		-1.227***		-1.239***
		(0.445)		(0.444)		(0.443)
Informal Competition (dummy)		-0.784*		-0.789*		-0.785*
		(0.445)		(0.446)		(0.446)
Firm Bought Assets (dummy)		3.095***		3.095***		3.096***
		(0.448)		(0.448)		(0.448)
Crime Losses (% of sales)		-0.015		-0.015		-0.015
		(0.047)		(0.047)		(0.047)
Taxes: Obstacle Severity		-0.155		-0.156		-0.156

		(0.175)		(0.175)		(0.175)
Labor Laws: Obstacle Severity		0.021		0.020		0.016
		(0.166)		(0.166)		(0.166)
Industry fixed effects		Yes	Yes	Yes	Yes	Yes
Country fixed effects		Yes	Yes	Yes	Yes	Yes
Number of observations	41,197	41,183	41,194	41,183	41,193	41,183
R-squared	0.167	0.186	0.167	0.186	0.167	0.186

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). Sample size varies due to missing data.

Online Appendix

Table A1: Description of Variables

Variable name	Description of variable
Employment Growth	Total number of permanent full-time workers in the final year (last fiscal year) minus the same in the initial year (beginning of the growth period which is 2 fiscal years ago for most countries and 3 fiscal years ago for other countries) divided by the average (mean value) number of permanent workers in the final and initial year. The ratio is divided by the number of years between the final and initial year and multiplied by 100 to convert to percentage terms. Source: Enterprise Surveys. www.enterprisesurveys.org
Employment Growth 1	Log of the total number of permanent full-time workers at the firm in the final year (last fiscal year) minus the log of the total number of permanent full-time workers at the firm in the initial year (beginning of the growth period which is 2 fiscal years ago for most countries and 3 fiscal years ago for other countries). The log difference is then divided by the number of years between the initial and final year and the multiplied by 100 to covert to percentage terms. Source: Enterprise Surveys. www.enterprisesurveys.org
Overall Corruption (CI avg.)	The ES asked firms the amount of money firms like itself pay as a proportion of their annual sales to public officials to get "things done". We take the average of the reported bribe amount (as percentage of firms' annual sales) where the average if taken over all other firms (that is, excluding the firm in question) in the same country-industry cell. Industry grouping is defined at the 2-digit ISIC Rev. 3.1 level. There are 7 industries in our sample. Source: Enterprise Surveys. www.enterprisesurveys.org
Petty Corruption Incidence (CI avg.)	The ES asked firms if they were requested to pay a bribe or expected to do so in soliciting the following services: obtaining electricity connection, obtaining water connection,

	obtaining construction permit, obtaining import license, obtaining operating license, and inspections or meetings with tax officials. Based on this information, the ES compiles a measure of the incidence of petty corruption which is dummy variable equal to 1 if a firm experienced a bribe payment or request in one or more of the six transactions listed above and 0 otherwise. This incidence of petty corruption measure is available for only those firms that solicited the public services listed above or were inspected by tax officials. We use average value of the dummy variable obtained above, where the average is taken over all firms in the country-industry cell other than the firm in questions. Industry grouping is defined at the 2-digit ISIC Rev. 3.1 level. There are 7 industries in our sample. Source: Enterprise Surveys. www.enterprisesurveys.org
Petty Corruption Depth (CI avg.)	The ES asked firms if they were requested to pay a bribe or expected to do so in soliciting the following services: obtaining electricity connection, obtaining water connection, obtaining construction permit, obtaining import license, obtaining operating license, and inspections or meetings with tax officials. Based on this information, the ES compiles a measure of the depth of petty corruption which equals the proportion of the 6 transactions listed above for which the firm experienced a bribe payment or request for one. This corruption measure is available for only those firms that solicited the public services listed above or were inspected by tax officials. Following the ES methodology, a refusal to answer a question on whether bribes were requested or expected is considered as an affirmative answer. We use average value of the petty corruption depth variable as defined above, where the average is taken over all firms in the country-industry cell other than the firm in questions. Industry grouping is defined at the 2-digit ISIC

	Rev. 3.1 level. There are 7 industries in our sample. Source: Enterprise Surveys. www.enterprisesurveys.org
Financially Constrained (CI avg.)	The ES asked firms if they applied for a loan during the last fiscal year. If they did not, they were asked to choose the main reason for not doing so from the following list: No need for a loan as the firm has enough internal funds; application procedures were too complex; interest rates were not favorable; collateral requirements were too high; loan size and maturity was insufficient; or did not think it would be approved. For firms that did apply for a loan, the ES asks if the loan application was rejected, still pending, approved partially, or approved in full. Using this information, we first define a firm as financially unconstrained if it either did not apply for a loan because there was no need to do so (has enough internal funds) or if it applied for a loan and the loan amount was approved in full. The remaining firms are classified as financially constrained. Next, we compute the proportion of all firms in the same country-industry cell as the firm in question that are financially constrained. This proportion is computed overall all firms other than the firm in question. Industry grouping is defined at the 2-digit ISIC Rev. 3.1 level. There are 7 industries in our sample. Source: Enterprise Surveys. www.enterprisesurveys.org
Finance: A Major Obstacle (CI avg.)	The ES asked firms if (lack of) access to finance is no obstacle (0), minor obstacle (1), moderate obstacle (2), major obstacle (3), and a very severe obstacle (4) for the firms' day to day operations. We first define a dummy variable equal to 1 if the firm reported major or very severe obstacle and 0 otherwise. Next, we take the average of the dummy variable where the average if taken over all other firms (that is, excluding the firm in question) in the same country-industry cell. Industry grouping is defined at the 2-digit ISIC Rev. 3.1 level. There are 7 industries in our sample.

	Source: Enterprise Surveys. www.enterprisesurveys.org
Initial Employment (logs)	Log of total number of permanent full-time workers at the firm in the initial year (beginning of the growth period which is 2 fiscal years ago for most countries and 3 fiscal years ago for other countries). Source: Enterprise Surveys. www.enterprisesurveys.org
Age of Firm (logs)	Log of age of the firm. The age of the firm is defined as the year the Enterprise Survey was conducted minus the year the firm first started operations. Source: Enterprise Surveys. www.enterprisesurveys.org
Manager Experience (logs)	Log of the number of years of experience the top manager of the firm has working in the industry. Source: Enterprise Surveys. www.enterprisesurveys.org
Exports (proportion of sales)	The proportion of the firms' annual sales made abroad in the last fiscal year. Only direct exports are included. Source: Enterprise Surveys. www.enterprisesurveys.org
Foreign Ownership (% ownership)	Percentage of the firm that is owned by foreign individuals, companies or organizations. Source: Enterprise Surveys. www.enterprisesurveys.org
Female Manger (dummy)	Dummy variable equal to 1 if the top manager of the firm is a female and 0 otherwise. Source: Enterprise Surveys. www.enterprisesurveys.org
Multi-firm (dummy)	Dummy variable equal to 1 if the firm is part of a larger parent firm and 0 otherwise. Source: Enterprise Surveys. www.enterprisesurveys.org
Power outages (logs)	Log of (1 plus) total hours of power outages experienced by the firm in a typical month over the last fiscal year. Source: Enterprise Surveys. www.enterprisesurveys.org
Informal Competition (dummy)	Dummy variable equal to 1 if the firm reported competing against informal or unregistered firms and 0 otherwise.

	Source: Enterprise Surveys. www.enterprisesurveys.org
Purchased Bought Assets (dummy)	Dummy variable equal to 1 if the firm purchased fixed assets during the last fiscal year and 0 otherwise. Source: Enterprise Surveys. www.enterprisesurveys.org
Time Tax (%)	Percentage of the firms' senior management's time that is spent in dealing with requirements of government regulations. Source: Enterprise Surveys. www.enterprisesurveys.org
Courts: Obstacle Severity	Severity level of (lack of proper) functioning of courts as an obstacle to firms' day to day operations. The severity level is defined on a 0-4 scale as: no obstacle (0), minor obstacle (1), moderate obstacle (2), major obstacle (3), and a very severe obstacle (4). Source: Enterprise Surveys. www.enterprisesurveys.org
Crime Losses (% of sales)	Losses due to crime, theft, robbery, vandalism and arson during the last fiscal year and expressed as a percentage of the firms' annual sales. Source: Enterprise Surveys. www.enterprisesurveys.org
Taxes: Obstacle Severity	Severity level of (high) tax rates as an obstacle to firms' day to day operations. The severity level is defined on a 0-4 scale as: no obstacle (0), minor obstacle (1), moderate obstacle (2), major obstacle (3), and a very severe obstacle (4). Source: Enterprise Surveys. www.enterprisesurveys.org
Labor Laws: Obstacle Severity	Severity level of labor laws as an obstacle to firms' day to day operations. The severity level is defined on a 0-4 scale as: no obstacle (0), minor obstacle (1), moderate obstacle (2), major obstacle (3), and a very severe obstacle (4). Source: Enterprise Surveys. www.enterprisesurveys.org
Industry fixed effects	Dummy variables (one for each industry) indicating the industry to which the firm belongs. There are 7 industries defined at the 2-digit ISIC Rev. 3.1 level.

	Source: Enterprise Surveys. www.enterprisesurveys.org
Country fixed effects	Dummy variables (one for each country) indicating the country where the firm is located. Source: Enterprise Surveys. www.enterprisesurveys.org
GDP per capita (logs)	Log of GDP per capita (PPP adjusted and at constant 2011 International Dollars). Values of the variable used are for the fiscal year in which the Enterprise Survey was conducted. In cases where the Enterprise Survey spanned more than one calendar year, the GDP per capita value is the weighted average over the calendar year values covered by Enterprise Surveys, where the weights used are the proportion of firms surveyed in the calendar year. Source: World Development Indicators, World Bank.
Growth rate of GDP per capita	Annual percentage growth rate of real GDP per capita. Values of the variable used are for the fiscal year in which the Enterprise Survey was conducted. In cases where the Enterprise Survey spanned more than one calendar year, the GDP per capita growth rate value is the weighted average over the calendar year values covered by Enterprise Surveys, where the weights used are the proportion of firms surveyed in the calendar year. Source: World Development Indicators, World Bank.
Courts: Obstacle Severity (CI avg.)	The ES asked firms if (lack of proper) functioning of courts is no obstacle (0), minor obstacle (1), moderate obstacle (2), major obstacle (3), and a very severe obstacle (4) for the firms' day to day operations. We take the average of the reported obstacle severity level where the average if taken over all other firms (that is, excluding the firm in question) in the same country-industry cell. Industry grouping is defined at the 2-digit ISIC Rev. 3.1 level. There are 7 industries in our sample. Source: Enterprise Surveys. www.enterprisesurveys.org

Table A2: Summary Statistics of the main variables

	Mean	Std. deviation	Minimum	Maximum	Observations
Employment Growth	5.049	16.169	-94	98.7	41,203
Employment Growth 1	5.295	17.996	-170	251	41,203
Overall Corruption (CI avg.)	1.181	1.814	0	27.2	41,203
Petty Corruption Depth (CI avg.)	13.766	14.638	0	100	41,117
Petty Corruption Incidence (CI avg.)	17.510	16.902	0	100	41,117
Financially Constrained (CI avg.)	0.468	0.196	0	1	41,203
Initial Employment (logs)	2.447	0.883	0	4.6	41,203
Age of firm (logs)	2.578	0.678	0	5.4	41,203
Manager Experience (logs)	2.638	0.709	0	4.1	41,203
Exports (proportion of sales)	0.052	0.184	0	1	41,203
Foreign Ownership (% ownership)	7.799	24.924	0	100	41,203
Multi-firm (dummy)	0.184	0.387	0	1	41,203
Female Manager (dummy)	0.191	0.393	0	1	41,203
Power Outage (logs)	1.361	1.727	0	6.6	41,203
Informal Competition (dummy)	0.559	0.496	0	1	41,203
Firm Bought Assets (dummy)	0.393	0.488	0	1	41,203
Time Tax (%)	9.525	15.626	0	100	41,203
Courts: Obstacle Severity	0.947	1.224	0	4	41,203
Crime Losses (% of sales)	0.879	4.001	0	100	41,203
Taxes: Obstacle Severity	1.776	1.308	0	4	41,203
Labor Laws: Obstacle Severity	0.975	1.121	0	4	41,203

Sample used is the baseline sample used for Overall Corruption (Table 1). CI avg. denotes country-industry cell average excluding the own firm as described in section 2.

Table A3: Correlations between main explanatory variables

	Overall Corruption (CI avg.)	Petty Corruption Incidence (CI avg.)	Petty Corruption Depth (CI avg.)	Financially Constrained (CI avg.)
Financially Constrained (CI avg.)	0.136	0.152	0.147	1.000
Initial Employment (logs)	-0.044	0.009	0.012	-0.008
Age of firm (logs)	-0.046	-0.059	-0.047	-0.092
Manager Experience (logs)	-0.070	-0.097	-0.085	-0.088
Exports (proportion of sales)	-0.035	-0.013	-0.012	-0.045
Foreign Ownership (% ownership)	0.007	0.040	0.027	0.012
Multi-firm (dummy)	-0.024	0.027	0.026	-0.139
Female Manager (dummy)	-0.034	-0.007	-0.006	-0.036
Power Outage (logs)	0.179	0.227	0.229	0.170
Informal Competition (dummy)	0.074	0.044	0.026	0.059
Firm Bought Assets (dummy)	-0.022	-0.020	-0.028	0.016
Time Tax (%)	0.035	-0.006	-0.012	0.000
Courts: Obstacle Severity	0.074	0.084	0.071	0.024
Crime Losses (% of sales)	0.104	0.048	0.027	0.038
Taxes: Obstacle Severity	0.000	-0.017	-0.028	0.055
Labor Laws: Obstacle Severity	0.057	-0.005	-0.019	0.036

Sample used is the baseline sample as in Tables 1 and 2. CI avg. denotes Country-Industry cell average excluding the own firm as described in section 2.

Table A4: Correlations with external (macro-level) corruption indicators

Panel A: Spearman's rank correlation

	Overall Corruption	Petty Corruption Incidence	Petty Corruption Depth
Control of Corruption (Worldwide Governance Indicators; N=124)	0.544***	0.648***	0.633***
Corruption Perception Index (Transparency International; N=117))	0.437***	0.586***	0.549***
Corruption (ICRG; N=88)	0.208***	0.291***	0.303***

Panel B: Pearson's correlation

	Overall Corruption	Petty Corruption Incidence	Petty Corruption Depth
Control of Corruption (Worldwide Governance Indicators; N=124)	0.405***	0.564***	0.544***
Corruption Perception Index (Transparency International; N=117))	0.330***	0.524***	0.484***
Corruption (ICRG; N=88)	0.192*	0.248**	0.226**

Number of countries varies due to missing data on the corruption indicator. Baseline sample described in section 2 and used in Tables 1 and 2 is used.
The correlations are computed by first averaging firm-level responses on overall corruption, incidence and depth of petty corruption at the country-level.
All corruption indicators have been re-scaled so that higher values of all the indicators means higher corruption
Significance level is denoted by ***(1%), **(5%) and *(10%).

Table A5: Base Regression Results for Petty Corruption Incidence

Dependent variable: *Employment Growth*	(1)	(2)	(3)	(4)
Petty Corruption Incidence (CI avg.)*Financially Constrained (CI avg.)	-0.168**	-0.180***	-0.175***	-0.173***
	(0.066)	(0.064)	(0.065)	(0.065)
Petty Corruption Incidence (CI avg.)	0.097**	0.105***	0.107***	0.106***
	(0.038)	(0.037)	(0.037)	(0.037)
Financially Constrained (CI avg.)	3.004	3.528*	3.538*	3.546*
	(1.969)	(2.030)	(2.041)	(2.043)
Initial Employment (logs)	-5.763***	-5.622***	-5.889***	-5.889***
	(0.322)	(0.311)	(0.310)	(0.314)
Age of firm (logs)		-2.061***	-1.950***	-1.950***
		(0.295)	(0.291)	(0.291)
Manager Experience (logs)		0.019	-0.083	-0.068
		(0.273)	(0.274)	(0.274)
Exports (proportion of sales)		1.608	1.402	1.406
		(1.804)	(1.830)	(1.842)
Foreign Ownership (% ownership)		0.022***	0.021**	0.021**
		(0.008)	(0.008)	(0.008)
Multi-firm (dummy)		1.905***	1.845***	1.833***
		(0.639)	(0.648)	(0.645)
Female Manager (dummy)			-1.335***	-1.334***
			(0.436)	(0.435)
Power Outage (logs)			0.201*	0.211*
			(0.120)	(0.120)
Informal Competition (dummy)			-0.766*	-0.725*
			(0.428)	(0.421)
Firm Bought Assets (dummy)			3.107***	3.135***
			(0.412)	(0.423)
Time Tax (%)				0.005
				(0.013)
Courts: Obstacle Severity				-0.033
				(0.198)
Crime Losses (% of sales)				-0.024
				(0.045)
Taxes: Obstacle Severity				-0.172
				(0.164)
Labor Laws: Obstacle Severity				-0.004
				(0.171)
Industry fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes
Number of observations	41,117	41,117	41,117	41,117

R-squared	0.167	0.177	0.186	0.186

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A6: Base Regression Results for Petty Corruption Depth				
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**
Petty Corruption Depth (CI avg.)*Financially	-0.218***	-0.233***	-0.230***	-0.228***
Constrained (CI avg.)	(0.079)	(0.077)	(0.078)	(0.078)
Petty Corruption Depth (CI avg.)	0.117***	0.128***	0.131***	0.129***
	(0.044)	(0.044)	(0.043)	(0.044)
Financially Constrained (CI avg.)	3.076	3.616*	3.663*	3.674*
	(1.928)	(1.987)	(1.992)	(1.994)
Initial Employment (logs)	-5.764***	-5.621***	-5.888***	-5.888***
	(0.322)	(0.311)	(0.310)	(0.314)
Age of firm (logs)		-2.065***	-1.954***	-1.954***
		(0.296)	(0.291)	(0.291)
Manager Experience (logs)		0.015	-0.086	-0.072
		(0.273)	(0.274)	(0.274)
Exports (proportion of sales)		1.601	1.394	1.399
		(1.802)	(1.828)	(1.840)
Foreign Ownership (% ownership)		0.022***	0.021**	0.021**
		(0.008)	(0.008)	(0.008)
Multi-firm (dummy)		1.900***	1.841***	1.828***
		(0.639)	(0.648)	(0.645)
Female Manager (dummy)			-1.328***	-1.327***
			(0.436)	(0.435)
Power Outage (logs)			0.204*	0.213*
			(0.121)	(0.120)
Informal Competition (dummy)			-0.767*	-0.727*
			(0.428)	(0.421)
Firm Bought Assets (dummy)			3.106***	3.134***
			(0.412)	(0.423)
Time Tax (%)				0.005
				(0.013)
Courts: Obstacle Severity				-0.033
				(0.198)
Crime Losses (% of sales)				-0.025
				(0.045)
Taxes: Obstacle Severity				-0.171
				(0.164)
Labor Laws: Obstacle Severity				-0.003
				(0.171)
Industry fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes
Number of observations	41,117	41,117	41,117	41,117
R-squared	0.167	0.177	0.186	0.186

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A7: Interaction term controls and Overall Corruption					
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
Overall Corruption (CI avg.)*Financially Constrained (CI avg.)	**-1.160****	**-1.333*****	**-1.432*****	**-1.642*****	**-1.206****
	(0.478)	**(0.496)**	**(0.512)**	**(0.528)**	**(0.503)**
Overall Corruption (CI avg.)	1.008***	1.388**	1.536**	1.625**	1.974
	(0.322)	(0.578)	(0.595)	(0.631)	(1.395)
Financially Constrained (CI avg.)	-2.605	0.474	1.744	2.215	-27.505
	(5.018)	(6.296)	(6.195)	(6.261)	(18.112)
Initial Employment (logs)	-5.876***	-5.521***	-5.581***	-5.574***	-5.510***
	(0.328)	(0.739)	(0.751)	(0.755)	(0.776)
Age of firm (logs)	-2.030***	-1.748***	-1.710**	-1.759***	-1.822***
	(0.294)	(0.675)	(0.676)	(0.671)	(0.685)
Manager Experience (logs)	0.025	0.031	0.025	0.013	0.029
	(0.274)	(0.274)	(0.274)	(0.273)	(0.274)
Exports (proportion of sales)	1.072	1.024	4.684*	4.768**	4.524*
	(2.198)	(2.200)	(2.399)	(2.407)	(2.548)
Foreign ownership (% ownership)	0.022**	0.023**	-0.010	-0.010	-0.015
	(0.009)	(0.009)	(0.017)	(0.017)	(0.018)
Multi-firm (dummy)	1.741**	1.759**	1.796***	1.792***	1.902***
	(0.709)	(0.706)	(0.691)	(0.690)	(0.685)
Female Manager (dummy)	-1.227***	-1.225***	-1.223***	-1.221***	-1.218***
	(0.443)	(0.442)	(0.440)	(0.440)	(0.442)
Power Outage (logs)	0.201	0.197	0.194	0.362	0.253
	(0.122)	(0.123)	(0.123)	(0.273)	(0.263)
Informal Competition (dummy)	-0.846*	-0.835*	-0.864**	-0.740	-0.637
	(0.435)	(0.437)	(0.433)	(0.969)	(0.990)
Firm Bought Assets (dummy)	3.047***	3.055***	4.163***	4.099***	3.964***
	(0.455)	(0.455)	(0.918)	(0.930)	(0.953)
Time Tax (%)	0.004	0.004	0.004	-0.011	-0.008
	(0.014)	(0.014)	(0.014)	(0.028)	(0.028)
Courts: Obstacle Severity	0.011	0.007	0.005	0.069	0.052
	(0.213)	(0.212)	(0.213)	(0.394)	(0.401)
Crime Losses (% of sales)	-0.013	-0.011	-0.010	0.094	0.080
	(0.046)	(0.046)	(0.047)	(0.105)	(0.106)
Taxes: Obstacle Severity	-0.160	-0.159	-0.160	-0.151	-0.168
	(0.166)	(0.167)	(0.167)	(0.166)	(0.166)
Labor Laws: Obstacle Severity	0.004	0.003	0.012	0.004	0.003
	(0.174)	(0.174)	(0.173)	(0.173)	(0.174)
Square of Overall Corruption (CI avg.)	-0.021	-0.017	-0.015	-0.014	-0.016
	(0.017)	(0.017)	(0.016)	(0.013)	(0.014)
Square of Financially Constrained (CI avg.)	3.844	3.607	3.236	3.067	2.575
	(4.863)	(4.853)	(4.775)	(4.690)	(4.763)

Overall Corruption (CI avg.)*Initial Employment	-0.216	-0.199	-0.201	-0.189
	(0.150)	(0.155)	(0.157)	(0.157)
Financially Constrained (CI avg.)*Initial Employment	-0.253	-0.184	-0.185	-0.407
	(1.363)	(1.377)	(1.382)	(1.421)
Overall Corruption (CI avg.)*Age of Firm	0.080	0.054	0.116	0.118
	(0.139)	(0.147)	(0.141)	(0.143)
Financially Constrained (CI avg.)*Age of Firm	-0.790	-0.816	-0.809	-0.703
	(1.306)	(1.308)	(1.295)	(1.311)
Overall Corruption (CI avg.)*Exports		-0.150	-0.176	-0.207
		(0.662)	(0.669)	(0.679)
Financially Constrained (CI avg.)*Exports		-8.047	-8.162	-7.446
		(6.024)	(6.061)	(6.168)
Overall Corruption (CI avg.)*Foreign Ownership		0.003	0.002	0.002
		(0.006)	(0.006)	(0.006)
Financially Constrained (CI avg.)*Foreign Ownership		0.061*	0.061*	0.072**
		(0.034)	(0.034)	(0.035)
Overall Corruption (CI avg.)*Firm Bought Assets		-0.266*	-0.261	-0.253
		(0.161)	(0.164)	(0.164)
Financially Constrained (CI avg.)*Firm Bought Assets		-1.726	-1.665	-1.424
		(1.735)	(1.747)	(1.765)
Overall Corruption (CI avg.)*Power Outage			-0.088**	-0.100**
			(0.043)	(0.045)
Financially Constrained (CI avg.)*Power Outage			-0.029	0.242
			(0.525)	(0.508)
Overall Corruption (CI avg.)*Time Tax			0.003	0.004
			(0.006)	(0.006)
Financially Constrained (CI avg.)*Time Tax			0.022	0.014
			(0.057)	(0.057)
Overall Corruption (CI avg.)*Courts: Obstacle Severity			0.083	0.066
			(0.073)	(0.074)
Financially Constrained (CI avg.)*Courts: Obstacle Severity			-0.373	-0.264
			(0.737)	(0.752)
Overall Corruption (CI avg.)*Informal Competition			-0.200	-0.216
			(0.172)	(0.173)

Financially Constrained (CI avg.)*Informal Competition				0.284	0.147
				(1.814)	(1.844)
Overall Corruption (CI avg.)*Crime Losses				0.006	0.004
				(0.011)	(0.012)
Financially Constrained (CI avg.)*Crime Losses				-0.243	-0.208
				(0.218)	(0.220)
Overall Corruption (CI avg.)*GDP per capita (logs)					-0.067
					(0.132)
Financially Constrained (CI avg.)*GDP per capita (logs)					2.860
					(1.781)
Overall Corruption (CI avg.)*Growth rate of GDP per capita					0.006
					(0.015)
Financially Constrained (CI avg.)*Growth rate of GDP per capita					1.795***
					(0.545)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	41,203	41,203	41,203	41,203	41,108
R-squared	0.186	0.187	0.187	0.188	0.190

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A8: Interaction term controls and Petty Corruption Incidence					
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
Petty Corruption Incidence (CI avg.)*Financially Constrained (CI avg.)	**-0.191*** **	**-0.191*** **	**-0.210*** **	**-0.209*** **	**-0.210*** **
	(0.061)	**(0.061)**	**(0.063)**	**(0.065)**	**(0.071)**
Petty Corruption Incidence (CI avg.)	-0.020	-0.047	-0.017	-0.045	-0.456**
	(0.052)	(0.083)	(0.087)	(0.079)	(0.201)
Financially Constrained (CI avg.)	1.772	4.056	5.702	6.949	-3.266
	(4.810)	(6.228)	(6.131)	(6.163)	(16.698)
Initial Employment (logs)	-5.883***	-6.101***	-6.214***	-6.233***	-6.226***
	(0.313)	(0.783)	(0.791)	(0.794)	(0.811)
Age of firm (logs)	-1.959***	-1.517**	-1.484**	-1.512**	-1.593**
	(0.291)	(0.694)	(0.695)	(0.693)	(0.708)
Manager Experience (logs)	-0.082	-0.081	-0.093	-0.097	-0.050
	(0.273)	(0.274)	(0.273)	(0.270)	(0.271)
Exports (proportion of sales)	1.320	1.303	5.592**	5.443**	5.361**
	(1.833)	(1.833)	(2.361)	(2.330)	(2.428)
Foreign Ownership (% ownership)	0.021**	0.021**	-0.006	-0.006	-0.011
	(0.008)	(0.008)	(0.018)	(0.018)	(0.019)
Multi-firm (dummy)	1.853***	1.853***	1.908***	1.897***	1.991***
	(0.643)	(0.639)	(0.615)	(0.612)	(0.611)
Female Manager (dummy)	-1.376***	-1.363***	-1.361***	-1.376***	-1.378***
	(0.432)	(0.431)	(0.430)	(0.427)	(0.432)
Power Outage (logs)	0.218*	0.218*	0.218*	0.607**	0.500*
	(0.120)	(0.120)	(0.119)	(0.281)	(0.279)
Informal Competition (dummy)	-0.728*	-0.740*	-0.752*	-1.220	-1.107
	(0.421)	(0.425)	(0.420)	(1.009)	(1.033)
Firm Bought Assets (dummy)	3.125***	3.122***	4.571***	4.484***	4.404***
	(0.423)	(0.422)	(0.913)	(0.923)	(0.942)
Time Tax (%)	0.006	0.006	0.006	-0.000	-0.002
	(0.013)	(0.013)	(0.013)	(0.029)	(0.029)
Courts: Obstacle Severity	-0.041	-0.036	-0.042	-0.197	-0.214
	(0.199)	(0.196)	(0.195)	(0.390)	(0.401)
Crime Losses (% of sales)	-0.022	-0.022	-0.021	0.073	0.064
	(0.045)	(0.046)	(0.046)	(0.105)	(0.106)
Taxes: Obstacle Severity	-0.174	-0.178	-0.165	-0.175	-0.190
	(0.163)	(0.165)	(0.168)	(0.166)	(0.166)

Labor Laws: Obstacle Severity	0.003	0.006	0.010	0.005	0.005
	(0.170)	(0.172)	(0.171)	(0.168)	(0.169)
Square of Petty Corruption Incidence (CI avg.)	0.002***	0.002***	0.002***	0.002***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Square of Financially Constrained (CI avg.)	2.169	1.967	1.211	0.164	-0.327
	(4.659)	(4.664)	(4.624)	(4.587)	(4.657)
Petty Corruption Incidence (CI avg.)*Initial Employment		0.014	0.020	0.022	0.022
		(0.021)	(0.019)	(0.018)	(0.019)
Financially Constrained (CI avg.)*Initial Employment		-0.056	-0.056	-0.044	-0.128
		(1.313)	(1.346)	(1.348)	(1.376)
Petty Corruption Incidence (CI avg.)*Age of Firm		-0.005	-0.008	-0.006	-0.006
		(0.015)	(0.015)	(0.015)	(0.015)
Financially Constrained (CI avg.)*Age of Firm		-0.756	-0.693	-0.655	-0.525
		(1.288)	(1.291)	(1.279)	(1.301)
Petty Corruption Incidence (CI avg.)*Exports			-0.047	-0.046	-0.048
			(0.104)	(0.103)	(0.104)
Financially Constrained (CI avg.)*Exports			-7.956	-7.770	-7.409
			(5.341)	(5.287)	(5.455)
Petty Corruption Incidence (CI avg.)*Foreign Ownership			-0.000	-0.000	-0.000
			(0.000)	(0.000)	(0.000)
Financially Constrained (CI avg.)*Foreign Ownership			0.074**	0.071**	0.078**
			(0.032)	(0.032)	(0.033)
Petty Corruption Incidence (CI avg.)*Firm Bought Assets			-0.044*	-0.047**	-0.045*
			(0.023)	(0.024)	(0.024)
Financially Constrained (CI avg.)*Firm Bought Assets			-1.479	-1.236	-1.185
			(1.704)	(1.727)	(1.752)
Petty Corruption Incidence (CI avg.)*Power Outage				-0.025***	-0.023***
				(0.006)	(0.007)

Financially Constrained (CI avg.)*Power Outage	0.296	0.423
	(0.487)	(0.472)
Petty Corruption Incidence (CI avg.)*Time Tax	-0.000	-0.000
	(0.001)	(0.001)
Financially Constrained (CI avg.)*Time Tax	0.019	0.024
	(0.057)	(0.057)
Petty Corruption Incidence (CI avg.)*Courts: Obstacle Severity	0.028**	0.030**
	(0.011)	(0.011)
Financially Constrained (CI avg.)*Courts: Obstacle Severity	-0.725	-0.714
	(0.751)	(0.767)
Petty Corruption Incidence (CI avg.)*Informal Competition	0.037*	0.038*
	(0.022)	(0.022)
Financially Constrained (CI avg.)*Informal Competition	-0.444	-0.709
	(1.804)	(1.834)
Petty Corruption Incidence (CI avg.)*Crime Losses	0.002	0.002
	(0.002)	(0.002)
Financially Constrained (CI avg.)*Crime Losses	-0.273	-0.253
	(0.191)	(0.193)
Petty Corruption Incidence (CI avg.)*GDP per capita (logs)		0.045**
		(0.021)
Financially Constrained (CI avg.)*GDP per capita (logs)		0.776
		(1.623)
Petty Corruption Incidence (CI avg.)*Growth rate of GDP per capita		-0.001
		(0.004)

Financially Constrained (CI avg.)*Growth rate of GDP per capita					1.503*** (0.477)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	41,117	41,117	41,117	41,117	41,022
R-squared	0.187	0.187	0.189	0.191	0.193

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A9: Interaction term controls and Petty Corruption Depth					
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
Petty Corruption Depth (CI avg.)*Financially Constrained (CI avg.)	**-0.251*** **	**-0.250*** **	**-0.268*** **	**-0.266*** **	**-0.256*** **
	(0.074)	**(0.075)**	**(0.078)**	**(0.080)**	**(0.085)**
Petty Corruption Depth (CI avg.)	0.012	-0.015	0.016	-0.032	-0.365
	(0.058)	(0.107)	(0.113)	(0.096)	(0.229)
Financially Constrained (CI avg.)	2.169	4.203	5.821	7.137	-5.046
	(4.807)	(6.248)	(6.151)	(6.174)	(16.367)
Initial Employment (logs)	-5.878***	-6.047***	-6.155***	-6.192***	-6.196***
	(0.312)	(0.779)	(0.783)	(0.785)	(0.803)
Age of firm (logs)	-1.964***	-1.567**	-1.554**	-1.578**	-1.662**
	(0.292)	(0.686)	(0.687)	(0.685)	(0.700)
Manager Experience (logs)	-0.089	-0.087	-0.101	-0.102	-0.059
	(0.274)	(0.275)	(0.274)	(0.271)	(0.272)
Exports (proportion of sales)	1.364	1.347	5.865**	5.736**	5.639**
	(1.831)	(1.818)	(2.558)	(2.513)	(2.619)
Foreign Ownership (% ownership)	0.021**	0.021**	-0.008	-0.007	-0.012
	(0.008)	(0.008)	(0.018)	(0.018)	(0.019)
Multi-firm (dummy)	1.853***	1.854***	1.913***	1.898***	1.993***
	(0.642)	(0.637)	(0.610)	(0.605)	(0.605)
Female Manager (dummy)	-1.379***	-1.369***	-1.369***	-1.386***	-1.389***
	(0.432)	(0.429)	(0.428)	(0.425)	(0.429)
Power Outage (logs)	0.220*	0.220*	0.221*	0.538*	0.446
	(0.121)	(0.120)	(0.120)	(0.277)	(0.275)
Informal Competition (dummy)	-0.714*	-0.719*	-0.732*	-1.224	-1.131
	(0.422)	(0.424)	(0.419)	(1.005)	(1.028)
Firm Bought Assets (dummy)	3.126***	3.124***	4.548***	4.472***	4.414***
	(0.422)	(0.422)	(0.905)	(0.917)	(0.935)
Time Tax (%)	0.005	0.006	0.006	-0.003	-0.006
	(0.013)	(0.013)	(0.013)	(0.029)	(0.029)
Courts: Obstacle Severity	-0.037	-0.033	-0.040	-0.188	-0.196
	(0.199)	(0.195)	(0.194)	(0.390)	(0.400)
Crime Losses (% of sales)	-0.023	-0.023	-0.022	0.079	0.070
	(0.046)	(0.046)	(0.046)	(0.104)	(0.105)
Taxes: Obstacle Severity	-0.176	-0.179	-0.165	-0.178	-0.192
	(0.163)	(0.165)	(0.169)	(0.167)	(0.167)

Labor Laws: Obstacle Severity	0.000	0.003	0.003	0.003	0.002
	(0.171)	(0.173)	(0.172)	(0.168)	(0.169)
Square of Petty Corruption Depth (CI avg.)	0.002***	0.002***	0.002***	0.002***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Square of Financially Constrained (CI avg.)	1.858	1.717	0.992	-0.012	-0.497
	(4.652)	(4.663)	(4.614)	(4.567)	(4.644)
Petty Corruption Depth (CI avg.)*Initial Employment		0.011	0.017	0.020	0.020
		(0.027)	(0.023)	(0.022)	(0.023)
Financially Constrained (CI avg.)*Initial Employment		0.047	0.066	0.120	0.031
		(1.316)	(1.346)	(1.349)	(1.379)
Petty Corruption Depth (CI avg.)*Age of Firm		-0.002	-0.004	-0.001	-0.001
		(0.016)	(0.017)	(0.017)	(0.017)
Financially Constrained (CI avg.)*Age of Firm		-0.787	-0.719	-0.704	-0.574
		(1.290)	(1.294)	(1.281)	(1.303)
Petty Corruption Depth (CI avg.)*Exports			-0.067	-0.064	-0.064
			(0.147)	(0.145)	(0.145)
Financially Constrained (CI avg.)*Exports			-8.189	-8.097	-7.773
			(5.363)	(5.292)	(5.485)
Petty Corruption Depth (CI avg.)*Foreign Ownership			-0.000	-0.000	-0.000
			(0.000)	(0.000)	(0.000)
Financially Constrained (CI avg.)*Foreign Ownership			0.071**	0.068**	0.076**
			(0.032)	(0.032)	(0.033)
Petty Corruption Depth (CI avg.)*Firm Bought Assets			-0.047*	-0.049*	-0.047
			(0.028)	(0.030)	(0.029)
Financially Constrained (CI avg.)*Firm Bought Assets			-1.713	-1.505	-1.465
			(1.701)	(1.720)	(1.745)
Petty Corruption Depth (CI avg.)*Power Outage				-0.026***	-0.024***
				(0.007)	(0.008)

Financially Constrained (CI avg.)*Power Outage	0.260	0.386
	(0.492)	(0.478)
Petty Corruption Depth (CI avg.)*Time Tax	0.000	-0.000
	(0.001)	(0.001)
Financially Constrained (CI avg.)*Time Tax	0.017	0.023
	(0.057)	(0.057)
Petty Corruption Depth (CI avg.)*Courts: Obstacle Severity	0.037**	0.038***
	(0.014)	(0.015)
Financially Constrained (CI avg.)*Courts: Obstacle Severity	-0.764	-0.749
	(0.753)	(0.770)
Petty Corruption Depth (CI avg.)*Informal Competition	0.045*	0.047*
	(0.026)	(0.026)
Financially Constrained (CI avg.)*Informal Competition	-0.352	-0.564
	(1.796)	(1.822)
Petty Corruption Depth (CI avg.)*Crime Losses	0.002	0.001
	(0.003)	(0.003)
Financially Constrained (CI avg.)*Crime Losses	-0.262	-0.240
	(0.185)	(0.187)
Petty Corruption Depth (CI avg.)*GDP per capita (logs)		0.036
		(0.024)
Financially Constrained (CI avg.)*GDP per capita (logs)		0.995
		(1.596)
Petty Corruption Depth (CI avg.)*Growth rate of GDP per capita		-0.000
		(0.005)
Financially Constrained (CI avg.)*Growth rate of GDP per capita		1.461***
		(0.475)

Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes
Constant	22.886***	22.256***	21.495***	21.747***	22.274***
	(2.166)	(3.138)	(3.146)	(3.166)	(3.293)
Number of observations	41,117	41,117	41,117	41,117	41,022
R-squared	0.187	0.187	0.188	0.191	0.192

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A10: Alternative measure of financial constraint

Dependent variable: *Employment Growth*	(1)	(2)	(3)	(4)
Overall Corruption (CI avg.)*Finance: A Major Obstacle (CI avg.)	-1.054***	-1.018***	-0.979***	-0.969***
	(0.328)	(0.314)	(0.334)	(0.332)
Overall Corruption (CI avg.)	0.488***	0.464***	0.465***	0.457***
	(0.177)	(0.170)	(0.173)	(0.172)
Finance: A Major Obstacle (CI avg.)	1.025	0.572	0.622	0.598
	(1.931)	(1.943)	(1.959)	(1.956)
Initial Employment (logs)	-5.766***	-5.620***	-5.885***	-5.885***
	(0.347)	(0.326)	(0.322)	(0.328)
Age of firm (logs)		-2.150***	-2.045***	-2.046***
		(0.295)	(0.293)	(0.293)
Manager Experience (logs)		0.107	0.013	0.024
		(0.272)	(0.274)	(0.273)
Exports (proportion of sales)		1.433	1.198	1.203
		(2.150)	(2.179)	(2.188)
Foreign Ownership (% ownership)		0.025***	0.023***	0.023***
		(0.009)	(0.009)	(0.009)
Multi-firm (dummy)		1.855***	1.780**	1.768**
		(0.704)	(0.714)	(0.708)
Female Manager (dummy)			-1.263***	-1.262***
			(0.445)	(0.444)
Power Outage (logs)			0.203	0.211*
			(0.125)	(0.123)
Informal Competition (dummy)			-0.830*	-0.800*
			(0.448)	(0.435)
Firm Bought Assets (dummy)			3.027***	3.050***
			(0.440)	(0.454)
Time Tax (%)				0.004
				(0.014)
Courts: Obstacle Severity				0.007
				(0.213)
Crime Losses (% of sales)				-0.016
				(0.046)
Taxes: Obstacle Severity				-0.165
				(0.167)
Labor Laws: Obstacle Severity				-0.001
				(0.174)
Industry fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes
Number of observations	41,227	41,227	41,227	41,227

R-squared	0.167	0.177	0.186	0.186

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A11: Alternative measure of financial constraints

	Petty Corruption Incidence				Petty Corruption Depth			
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Petty Corruption Incidence (CI avg.)*Finance: A Major Obstacle (CI avg.)	-0.172**	-0.184**	-0.186**	-0.188**				
	(0.077)	(0.076)	(0.076)	(0.076)				
Petty Corruption Incidence (CI avg.)	0.062**	0.068***	0.072***	0.072***				
	(0.026)	(0.026)	(0.026)	(0.026)				
Petty Corruption Depth (CI avg.)*Finance: A Major Obstacle (CI avg.)					-0.181**	-0.201**	-0.205**	-0.207**
					(0.090)	(0.089)	(0.088)	(0.088)
Petty Corruption Depth (CI avg.)					0.059**	0.067**	0.073**	0.073**
					(0.030)	(0.029)	(0.030)	(0.030)
Finance: A Major Obstacle (CI avg.)	1.643	1.546	1.501	1.510	1.004	0.961	0.945	0.949
	(2.368)	(2.404)	(2.425)	(2.428)	(2.243)	(2.284)	(2.299)	(2.299)
Initial Employment (logs)	-5.776***	-5.634***	-5.900***	-5.900***	-5.775***	-5.633***	-5.898***	-5.898***
	(0.323)	(0.312)	(0.310)	(0.315)	(0.322)	(0.312)	(0.310)	(0.315)
Age of firm (logs)		-2.063***	-1.953***	-1.953***		-2.066***	-1.956***	-1.956***
		(0.296)	(0.292)	(0.292)		(0.296)	(0.292)	(0.292)
Manager Experience (logs)		0.021	-0.082	-0.067		0.022	-0.081	-0.066
		(0.273)	(0.274)	(0.274)		(0.274)	(0.275)	(0.274)
Exports (proportion of sales)		1.597	1.394	1.398		1.597	1.393	1.397
		(1.806)	(1.831)	(1.843)		(1.806)	(1.831)	(1.843)
Foreign Ownership (% ownership)		0.022***	0.021**	0.021**		0.022***	0.021**	0.021**
		(0.008)	(0.008)	(0.008)		(0.008)	(0.008)	(0.008)
Multi-firm (dummy)		1.882***	1.821***	1.808***		1.888***	1.827***	1.815***
		(0.640)	(0.649)	(0.646)		(0.640)	(0.649)	(0.646)
Female Manager (dummy)			-1.363***	-1.362***			-1.361***	-1.360***
			(0.436)	(0.435)			(0.436)	(0.435)

Power Outage (logs)			0.201*	0.211*			0.203*	0.213*
			(0.120)	(0.119)			(0.121)	(0.120)
Informal Competition (dummy)			-0.740*	-0.695*			-0.736*	-0.690
			(0.428)	(0.422)			(0.428)	(0.422)
Firm Bought Assets (dummy)			3.106***	3.137***			3.106***	3.137***
			(0.412)	(0.423)			(0.412)	(0.423)
Time Tax (%)				0.006				0.006
				(0.013)				(0.013)
Courts: Obstacle Severity				-0.040				-0.041
				(0.198)				(0.198)
Crime Losses (% of sales)				-0.027				-0.027
				(0.045)				(0.045)
Taxes: Obstacle Severity				-0.178				-0.177
				(0.164)				(0.164)
Labor Laws: Obstacle Severity				-0.016				-0.017
				(0.171)				(0.171)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	41,130	41,130	41,130	41,130	41,130	41,130	41,130	41,130
R-squared	0.167	0.177	0.186	0.186	0.167	0.177	0.186	0.186

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A12: Alternative measure of employment growth

Dependent variable: *Employment Growth* 1	(1)	(2)	(3)	(4)
Overall Corruption (CI avg.)*Financially Constrained (CI avg.)	-1.509***	-1.498***	-1.398***	-1.397***
	(0.505)	(0.485)	(0.489)	(0.489)
Overall Corruption (CI avg.)	0.895**	0.881**	0.842**	0.837**
	(0.354)	(0.342)	(0.345)	(0.345)
Financially Constrained (CI avg.)	1.397	1.537	1.492	1.493
	(1.902)	(1.958)	(1.966)	(1.974)
Initial Employment (logs)	-6.444***	-6.298***	-6.572***	-6.574***
	(0.434)	(0.398)	(0.389)	(0.396)
Age of firm (logs)		-2.189***	-2.080***	-2.082***
		(0.327)	(0.325)	(0.325)
Manager Experience (logs)		0.118	0.022	0.031
		(0.298)	(0.300)	(0.299)
Exports (proportion of sales)		0.948	0.711	0.714
		(2.823)	(2.848)	(2.857)
Foreign Ownership (% ownership)		0.030***	0.029***	0.029***
		(0.010)	(0.011)	(0.011)
Multi-firm (dummy)		1.860**	1.783**	1.773**
		(0.889)	(0.895)	(0.885)
Female Manager (dummy)			-1.288**	-1.288***
			(0.499)	(0.498)
Power Outage (logs)			0.205	0.210
			(0.136)	(0.133)
Informal Competition (dummy)			-0.877*	-0.858*
			(0.526)	(0.506)
Firm Bought Assets (dummy)			3.135***	3.150***
			(0.523)	(0.546)
Time Tax (%)				0.005
				(0.017)
Courts: Obstacle Severity				0.051
				(0.252)
Crime Losses (% of sales)				-0.008
				(0.050)
Taxes: Obstacle Severity				-0.161
				(0.190)
Labor Laws: Obstacle Severity				-0.004
				(0.198)
Industry fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes

Number of observations	41,203	41,203	41,203	41,203
R-squared	0.165	0.174	0.181	0.182

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A13: Alternative measure of employment growth

	Petty Corruption Incidence				Petty Corruption Depth			
Dependent variable: *Employment Growth* 1	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Petty Corruption Incidence (CI avg.)*Financially Constrained (CI avg.)	**-0.196*** **	**-0.209*** **	**-0.204*** **	**-0.203*** **				
	(0.074)	**(0.073)**	**(0.074)**	**(0.074)**				
Petty Corruption Incidence (CI avg.)	0.113***	0.122***	0.124***	0.123***				
	(0.042)	(0.042)	(0.042)	(0.042)				
Petty Corruption Depth (CI avg.)*Financially Constrained (CI avg.)					**-0.256*** **	**-0.272*** **	**-0.268*** **	**-0.267*** **
					(0.088)	**(0.087)**	**(0.088)**	**(0.088)**
Petty Corruption Depth (CI avg.)					0.138***	0.150***	0.153***	0.151***
					(0.049)	(0.049)	(0.049)	(0.050)
Financially Constrained (CI avg.)	3.652*	4.201*	4.210*	4.214*	3.751*	4.315*	4.362*	4.369*
	(2.200)	(2.268)	(2.281)	(2.282)	(2.158)	(2.222)	(2.229)	(2.230)
Initial Employment (logs)	-6.440***	-6.305***	-6.583***	-6.585***	-6.440***	-6.304***	-6.582***	-6.584***
	(0.394)	(0.374)	(0.369)	(0.374)	(0.394)	(0.374)	(0.369)	(0.374)
Age of firm (logs)		-2.123***	-2.007***	-2.007***		-2.128***	-2.012***	-2.013***
		(0.328)	(0.323)	(0.323)		(0.328)	(0.323)	(0.323)
Manager Experience (logs)		0.027	-0.080	-0.067		0.022	-0.085	-0.072
		(0.298)	(0.299)	(0.299)		(0.298)	(0.300)	(0.299)
Exports (proportion of sales)		1.332	1.117	1.120		1.323	1.108	1.112
		(2.338)	(2.361)	(2.375)		(2.336)	(2.358)	(2.372)
Foreign Ownership (% ownership)		0.028***	0.027***	0.027***		0.028***	0.027***	0.027***
		(0.010)	(0.010)	(0.010)		(0.010)	(0.010)	(0.010)
Multi-firm (dummy)		1.986**	1.924**	1.912**		1.981**	1.920**	1.908**
		(0.792)	(0.799)	(0.792)		(0.792)	(0.798)	(0.792)
Female Manager (dummy)			-1.427***	-1.427***			-1.420***	-1.420***
			(0.483)	(0.482)			(0.483)	(0.482)

Power Outage (logs)			0.215	0.223*			0.217*	0.226*
			(0.131)	(0.129)			(0.131)	(0.129)
Informal Competition (dummy)			-0.781	-0.746			-0.782	-0.747
			(0.494)	(0.483)			(0.494)	(0.483)
Firm Bought Assets (dummy)			3.245***	3.269***			3.245***	3.269***
			(0.482)	(0.499)			(0.481)	(0.499)
Time Tax (%)				0.006				0.006
				(0.015)				(0.015)
Courts: Obstacle Severity				-0.002				-0.001
				(0.230)				(0.230)
Crime Losses (% of sales)				-0.021				-0.021
				(0.049)				(0.049)
Taxes: Obstacle Severity				-0.174				-0.174
				(0.186)				(0.186)
Labor Laws: Obstacle Severity				-0.009				-0.008
				(0.193)				(0.193)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	41,117	41,117	41,117	41,117	41,117	41,117	41,117	41,117
R-squared	0.164	0.173	0.181	0.181	0.164	0.173	0.181	0.181

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A14: Corruption-Finance-Institutional Quality nexus (for Overall Corruption)				
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**
Overall Corruption (CI avg.)*Financially Constrained (CI avg.)*Courts: Obstacle Severity (CI avg.)	**-1.526*** **	**-1.522*** **	**-1.502*** **	**-1.507*** **
	(0.553)	**(0.546)**	**(0.541)**	**(0.539)**
Overall Corruption (CI avg.)*Financially Constrained (CI avg.)	0.543	0.571	0.623	0.633
	(0.777)	(0.772)	(0.789)	(0.787)
Financially Constrained (CI avg.)*Courts: Obstacle Severity (CI avg.)	-0.801	-0.550	-0.476	-0.469
	(2.461)	(2.455)	(2.451)	(2.456)
Overall Corruption (CI avg.)*Courts: Obstacle Severity (CI avg.)	1.021***	1.006***	1.000***	1.004***
	(0.343)	(0.342)	(0.343)	(0.340)
Overall Corruption (CI avg.)	-0.281	-0.295	-0.312	-0.323
	(0.414)	(0.412)	(0.428)	(0.426)
Financially Constrained (CI avg.)	2.097	2.052	1.961	1.964
	(2.752)	(2.800)	(2.832)	(2.837)
Courts: Obstacle Severity (CI avg.)	-0.255	-0.664	-0.717	-0.731
	(1.435)	(1.451)	(1.455)	(1.455)
Initial Employment (logs)	-5.754***	-5.601***	-5.866***	-5.867***
	(0.348)	(0.325)	(0.322)	(0.328)
Age of firm (logs)		-2.145***	-2.039***	-2.041***
		(0.295)	(0.293)	(0.293)
Manager Experience (logs)		0.120	0.029	0.040
		(0.273)	(0.275)	(0.274)
Exports (proportion of sales)		1.301	1.069	1.071
		(2.153)	(2.185)	(2.198)
Foreign Ownership (% ownership)		0.024***	0.022**	0.022**
		(0.009)	(0.009)	(0.009)
Multi-firm (dummy)		1.841***	1.768**	1.756**
		(0.705)	(0.715)	(0.712)
Female Manager (dummy)			-1.229***	-1.229***
			(0.444)	(0.443)
Power Outage (logs)			0.181	0.188

			(0.125)	(0.123)
Informal Competition (dummy)			-0.868*	-0.841*
			(0.449)	(0.440)
Firm Bought Assets (dummy)			3.034***	3.053***
			(0.441)	(0.454)
Time Tax (%)				0.005
				(0.015)
Crime Losses (% of sales)				-0.011
				(0.046)
Taxes: Obstacle Severity				-0.161
				(0.172)
Labor Laws: Obstacle Severity				0.009
				(0.166)
Industry fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes
Constant	19.659***	24.013***	23.597***	23.830***
	(2.329)	(2.455)	(2.453)	(2.385)
Number of observations	41,197	41,197	41,197	41,197
R-squared	0.168	0.177	0.186	0.187

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).

Table A15: Corruption-Finance-Institutional Quality nexus (for Petty corruption)

	Petty Corruption Incidence				Petty Corruption Depth			
Dependent variable: *Employment Growth*	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Petty Corruption Incidence (CI avg.)*Financially Constrained (CI avg.)*Courts: Obstacle Severity (CI avg.)	**-0.237**** **(0.103)**	**-0.217**** **(0.101)**	**-0.188*** **(0.102)**	**-0.186*** **(0.102)**				
Petty Corruption Depth (CI avg.)*Financially Constrained (CI avg.)*Courts: Obstacle Severity (CI avg.)					**-0.328***** **(0.119)**	**-0.301***** **(0.116)**	**-0.271**** **(0.117)**	**-0.270**** **(0.116)**
Petty Corruption Incidence (CI avg.)*Financially Constrained (CI avg.)	0.131 (0.134)	0.093 (0.131)	0.063 (0.133)	0.062 (0.132)				
Petty Corruption Incidence (CI avg.)*Courts: Obstacle Severity (CI avg.)	0.096* (0.053)	0.083 (0.052)	0.066 (0.052)	0.065 (0.052)				
Petty Corruption Incidence (CI avg.)	-0.030 (0.074)	-0.005 (0.074)	0.016 (0.074)	0.017 (0.074)				

Petty Corruption Depth (CI avg.)*Financially Constrained (CI avg.)					0.176	0.126	0.095	0.094
					(0.156)	(0.153)	(0.156)	(0.155)
Financially Constrained (CI avg.)*Courts: Obstacle Severity (CI avg.)	0.968	1.163	0.899	0.880	1.115	1.269	1.131	1.128
	(2.932)	(2.916)	(2.924)	(2.939)	(2.845)	(2.827)	(2.830)	(2.843)
Petty Corruption Depth (CI avg.)*Courts: Obstacle Severity (CI avg.)					0.146**	0.128**	0.110*	0.109*
					(0.060)	(0.060)	(0.059)	(0.059)
Petty Corruption Depth (CI avg.)					-0.068	-0.035	-0.011	-0.011
					(0.086)	(0.086)	(0.088)	(0.087)
Financially Constrained (CI avg.)	1.756	2.236	2.496	2.522	1.804	2.322	2.493	2.505
	(3.373)	(3.409)	(3.451)	(3.465)	(3.300)	(3.338)	(3.382)	(3.396)
Courts: Obstacle Severity (CI avg.)	-0.535	-0.866	-0.739	-0.736	-0.746	-1.066	-0.974	-0.977
	(1.549)	(1.569)	(1.574)	(1.576)	(1.512)	(1.529)	(1.533)	(1.535)
Initial Employment (logs)	-5.758***	-5.616***	-5.881***	-5.883***	-5.764***	-5.620***	-5.884***	-5.886***
	(0.321)	(0.310)	(0.309)	(0.313)	(0.320)	(0.309)	(0.308)	(0.312)
Age of firm (logs)		-2.064***	-1.957***	-1.957***		-2.063***	-1.956***	-1.957***
		(0.295)	(0.291)	(0.291)		(0.295)	(0.291)	(0.291)
Manager Experience (logs)		0.027	-0.073	-0.060		0.025	-0.075	-0.062
		(0.272)	(0.273)	(0.273)		(0.273)	(0.274)	(0.273)
Exports (proportion of sales)		1.603	1.407	1.410		1.590	1.393	1.397
		(1.786)	(1.810)	(1.822)		(1.781)	(1.805)	(1.817)

Foreign Ownership (% ownership)		0.022***	0.021**	0.021**		0.022***	0.021**	0.021**
		(0.008)	(0.008)	(0.008)		(0.008)	(0.008)	(0.008)
Multi-firm (dummy)		1.892***	1.830***	1.818***		1.879***	1.818***	1.806***
		(0.639)	(0.648)	(0.645)		(0.639)	(0.648)	(0.646)
Female Manager (dummy)			-1.344***	-1.343***			-1.334***	-1.333***
			(0.435)	(0.434)			(0.434)	(0.433)
Power Outage (logs)			0.195	0.203*			0.195	0.204*
			(0.121)	(0.120)			(0.121)	(0.120)
Informal Competition			-0.741*	-0.706*			-0.749*	-0.714*
(dummy)			(0.429)	(0.425)			(0.429)	(0.424)
Firm Bought Assets			3.099***	3.122***			3.097***	3.121***
(dummy)			(0.411)	(0.421)			(0.410)	(0.420)
Time Tax (%)				0.006				0.006
				(0.013)				(0.013)
Crime Losses (% of sales)				-0.023				-0.024
				(0.045)				(0.045)
Taxes: Obstacle Severity				-0.176				-0.175
				(0.165)				(0.165)
Labor Laws: Obstacle Severity				-0.008				-0.005
				(0.165)				(0.165)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	41,108	41,108	41,108	41,108	41,108	41,108	41,108	41,108
R-squared	0.168	0.177	0.187	0.187	0.168	0.178	0.187	0.187

Standard errors in brackets. All standard errors are Huber-White robust and clustered on the country times industry level. Significance is denoted by *** (1%), ** (5%), * (10%). All regressions use a constant term (not shown).